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(54) ARTIFICIAL INTELLIGENCE ASSISTANT FOR VEHICLE DIAGNOSTICS

(71) Applicant: **INNOVA ELECTRONICS CORPORATION**, IRVINE, CA (US)

(72) Inventors: **Thuan Huynh**, Ho Chi Minh City (VN); **Danh Nguyen**, Ho Chi Minh City (VN); **Huy Mai**, Ho Chi Minh City (VN)

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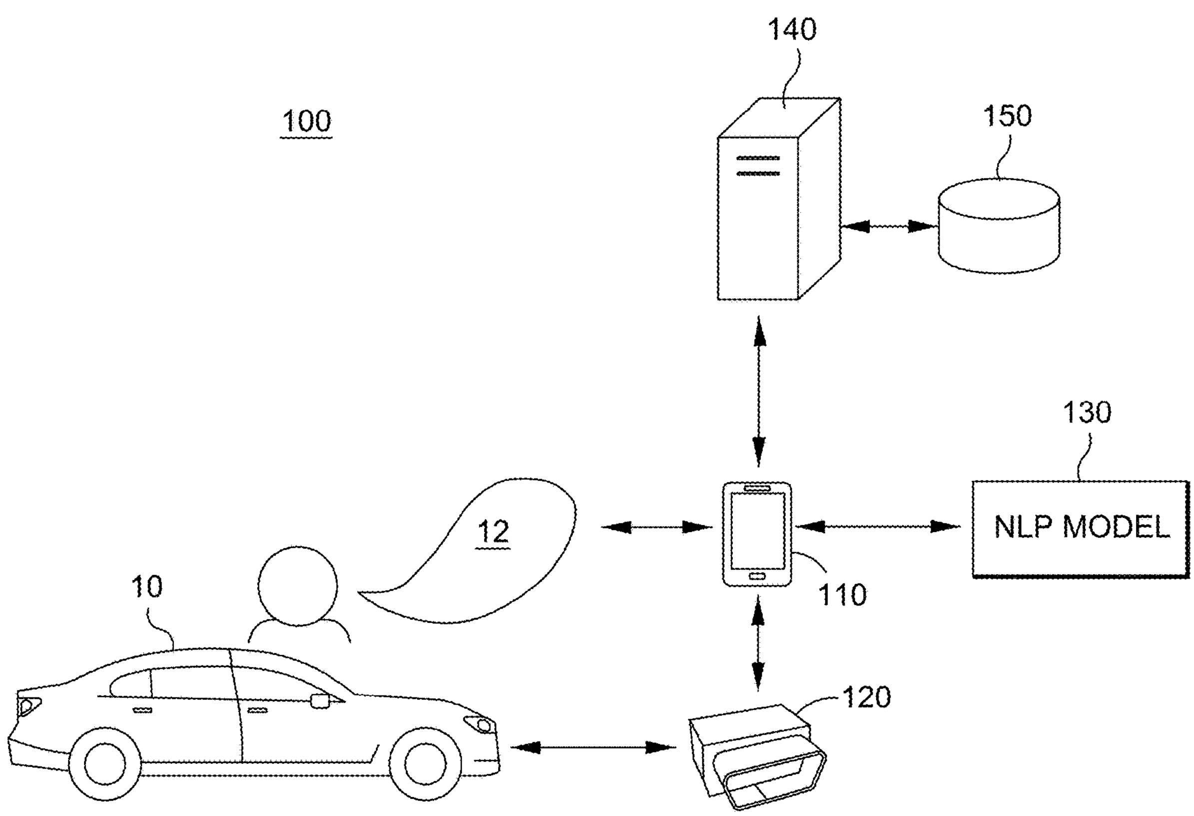
CPC ***G07C 5/0833*** (2013.01); ***G06F 40/40*** (2020.01); ***G07C 5/008*** (2013.01); ***G07C 5/0808*** (2013.01); *G07C 2205/02* (2013.01)

(57) **ABSTRACT**

A method of providing vehicle diagnostics includes receiving a request for AI-assisted diagnostics, receiving diagnostic data obtained from a vehicle, prompting a user to describe one or more symptoms exhibited by the vehicle, processing the one or more symptoms and the diagnostic data at least in part using a natural language processing (NLP) model, and providing natural language guidance to the user based on a result of the processing. The natural language guidance may include testing instructions for the user to perform a step-by-step procedure to diagnose the vehicle and/or repair instructions for the user to perform a repair on the vehicle.



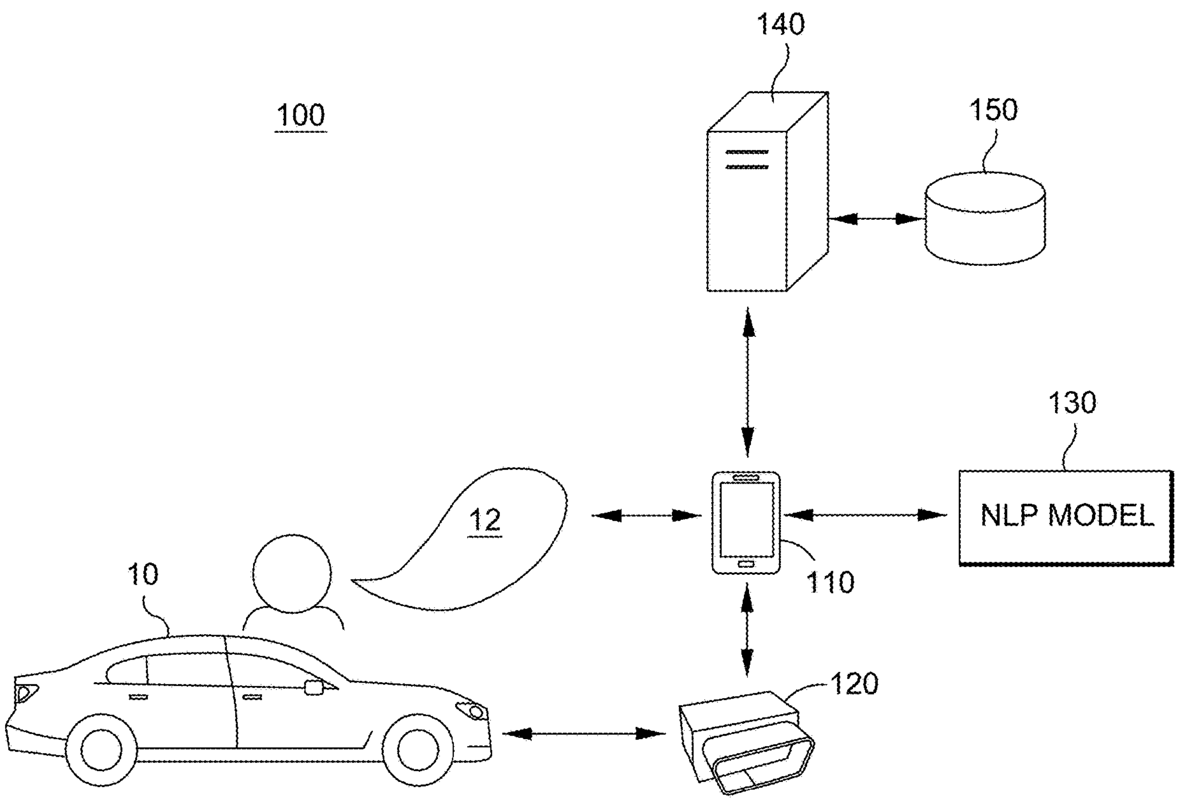


FIG. 1

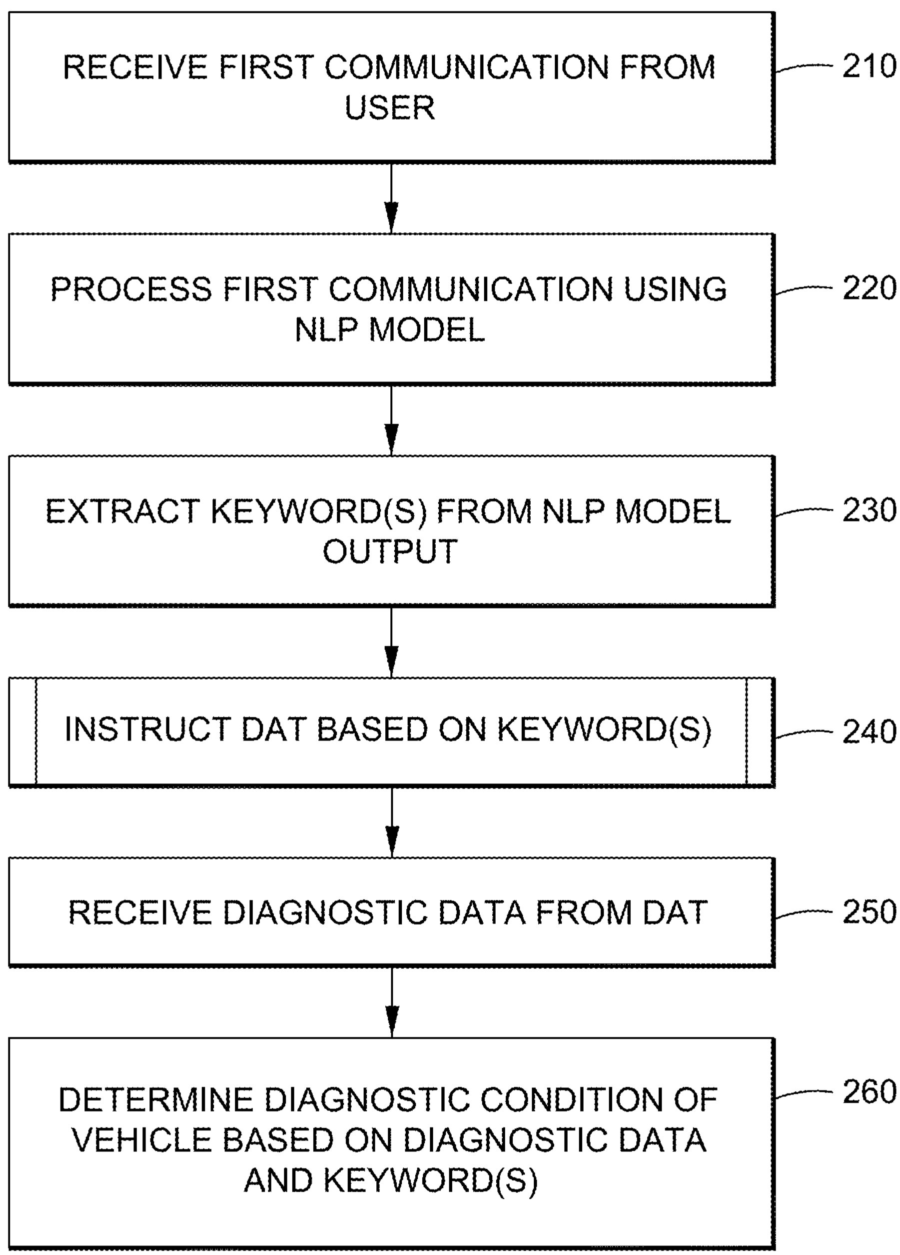

RECEIVE FIRST COMMUNICATION FROM USER
210
PROCESS FIRST COMMUNICATION USING NLP MODEL
220
EXTRACT KEYWORD(S) FROM NLP MODEL OUTPUT
230
INSTRUCT DAT BASED ON KEYWORD(S)
240
RECEIVE DIAGNOSTIC DATA FROM DAT
250
DETERMINE DIAGNOSTIC CONDITION OF VEHICLE BASED ON DIAGNOSTIC DATA AND KEYWORD(S)
260

FIG. 2

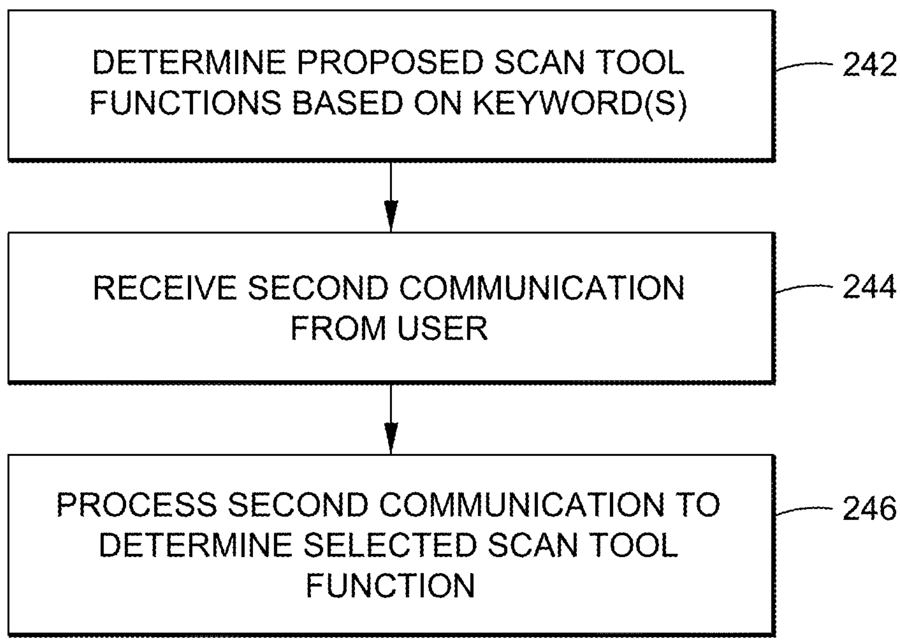

DETERMINE PROPOSED SCAN TOOL FUNCTIONS BASED ON KEYWORD(S)
242
RECEIVE SECOND COMMUNICATION FROM USER
244
PROCESS SECOND COMMUNICATION TO DETERMINE SELECTED SCAN TOOL FUNCTION
246

FIG. 3

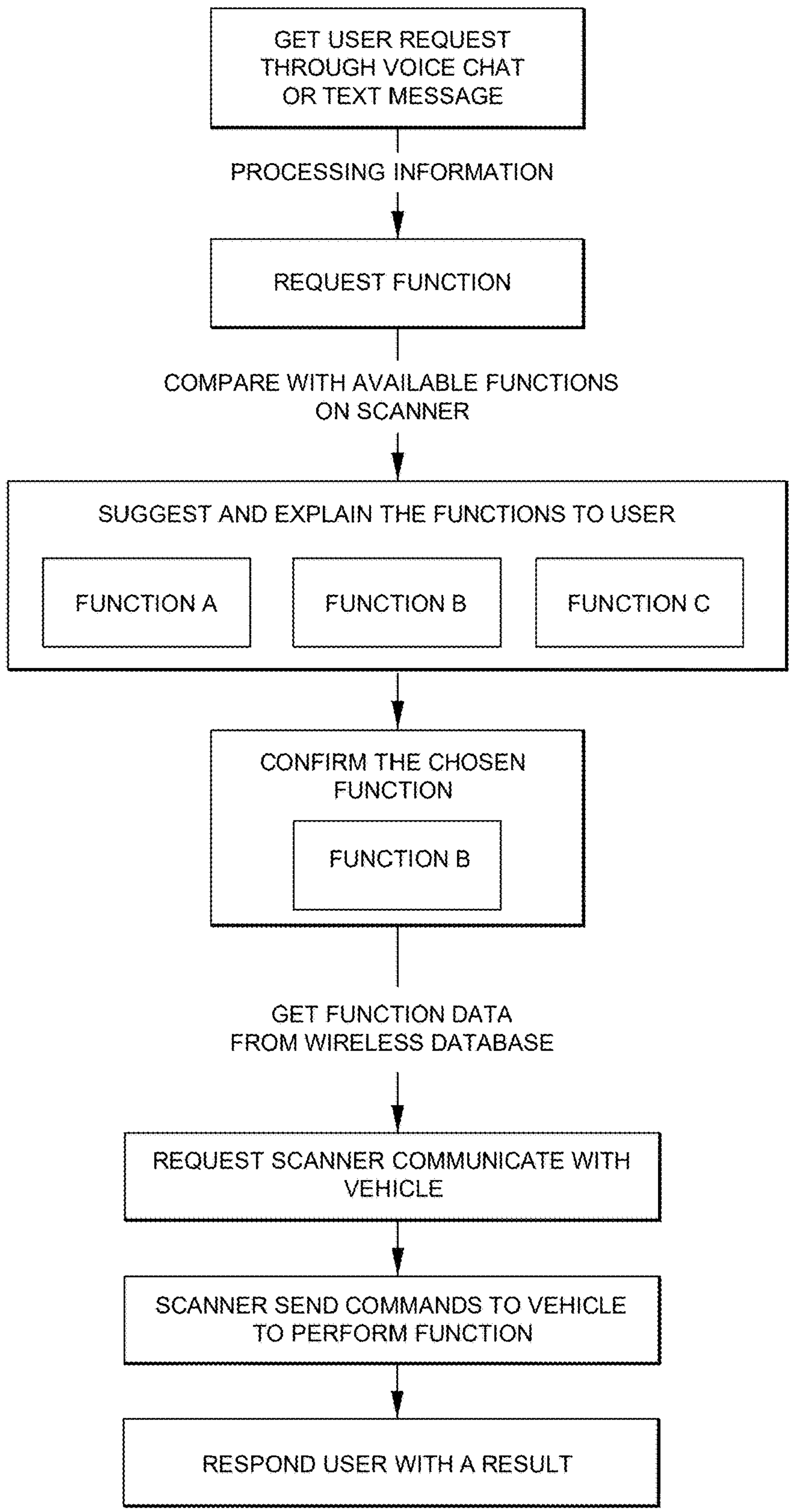

GET USER REQUEST THROUGH VOICE CHAT OR TEXT MESSAGE
PROCESSING INFORMATION
REQUEST FUNCTION
COMPARE WITH AVAILABLE FUNCTIONS ON SCANNER
SUGGEST AND EXPLAIN THE FUNCTIONS TO USER
FUNCTION A
FUNCTION B
FUNCTION C
CONFIRM THE CHOSEN FUNCTION
FUNCTION B
GET FUNCTION DATA FROM WIRELESS DATABASE
REQUEST SCANNER COMMUNICATE WITH VEHICLE
SCANNER SEND COMMANDS TO VEHICLE TO PERFORM FUNCTION
RESPOND USER WITH A RESULT

FIG. 4

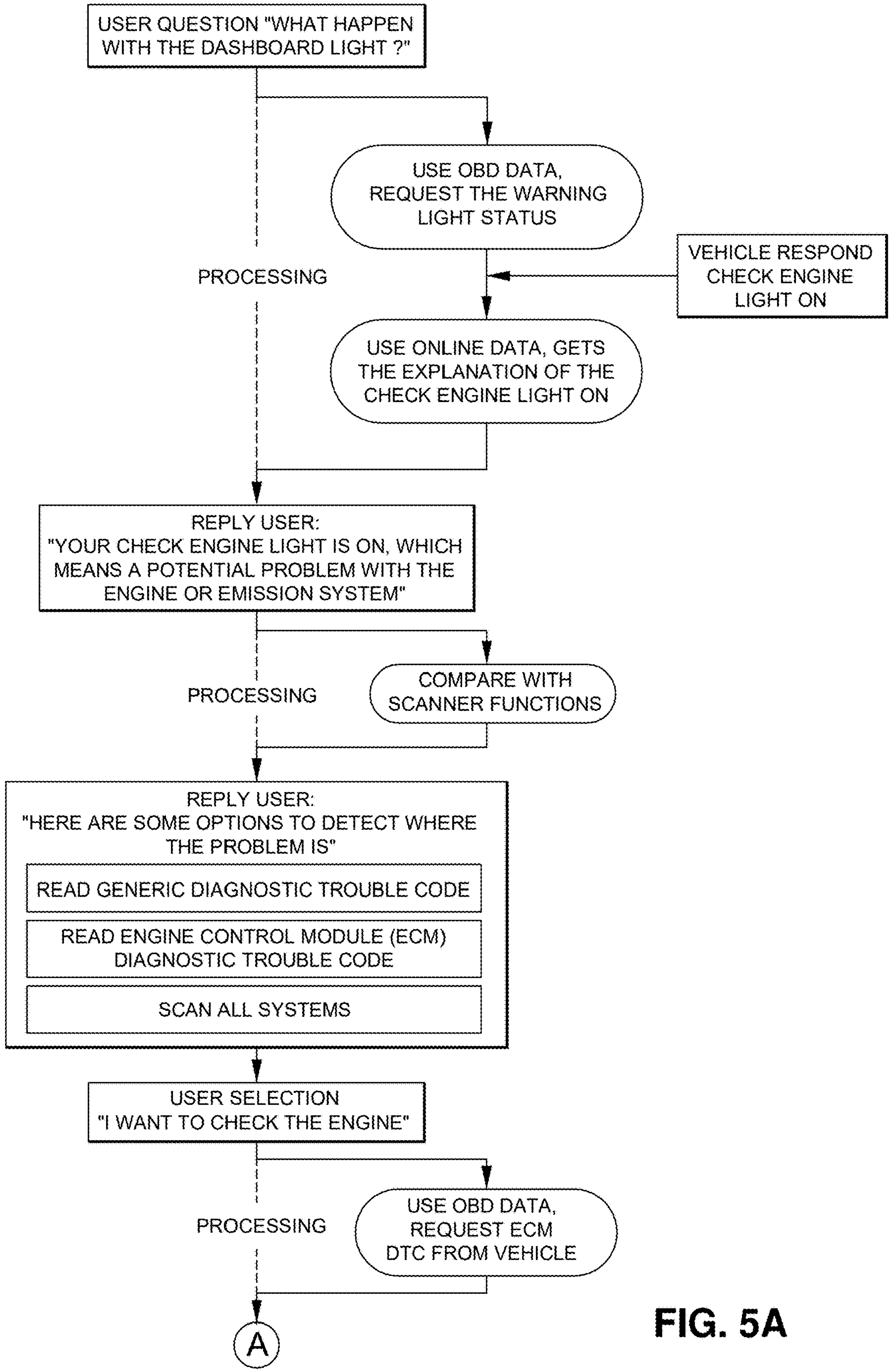

USER QUESTION "WHAT HAPPEN WITH THE DASHBOARD LIGHT ?"
USE OBD DATA, REQUEST THE WARNING LIGHT STATUS
VEHICLE RESPOND CHECK ENGINE LIGHT ON
PROCESSING
USE ONLINE DATA, GETS THE EXPLANATION OF THE CHECK ENGINE LIGHT ON
REPLY USER: "YOUR CHECK ENGINE LIGHT IS ON, WHICH MEANS A POTENTIAL PROBLEM WITH THE ENGINE OR EMISSION SYSTEM"
PROCESSING
COMPARE WITH SCANNER FUNCTIONS
REPLY USER: "HERE ARE SOME OPTIONS TO DETECT WHERE THE PROBLEM IS"
READ GENERIC DIAGNOSTIC TROUBLE CODE
READ ENGINE CONTROL MODULE (ECM) DIAGNOSTIC TROUBLE CODE
SCAN ALL SYSTEMS
USER SELECTION "I WANT TO CHECK THE ENGINE"
PROCESSING
USE OBD DATA, REQUEST ECM DTC FROM VEHICLE
A

FIG. 5A

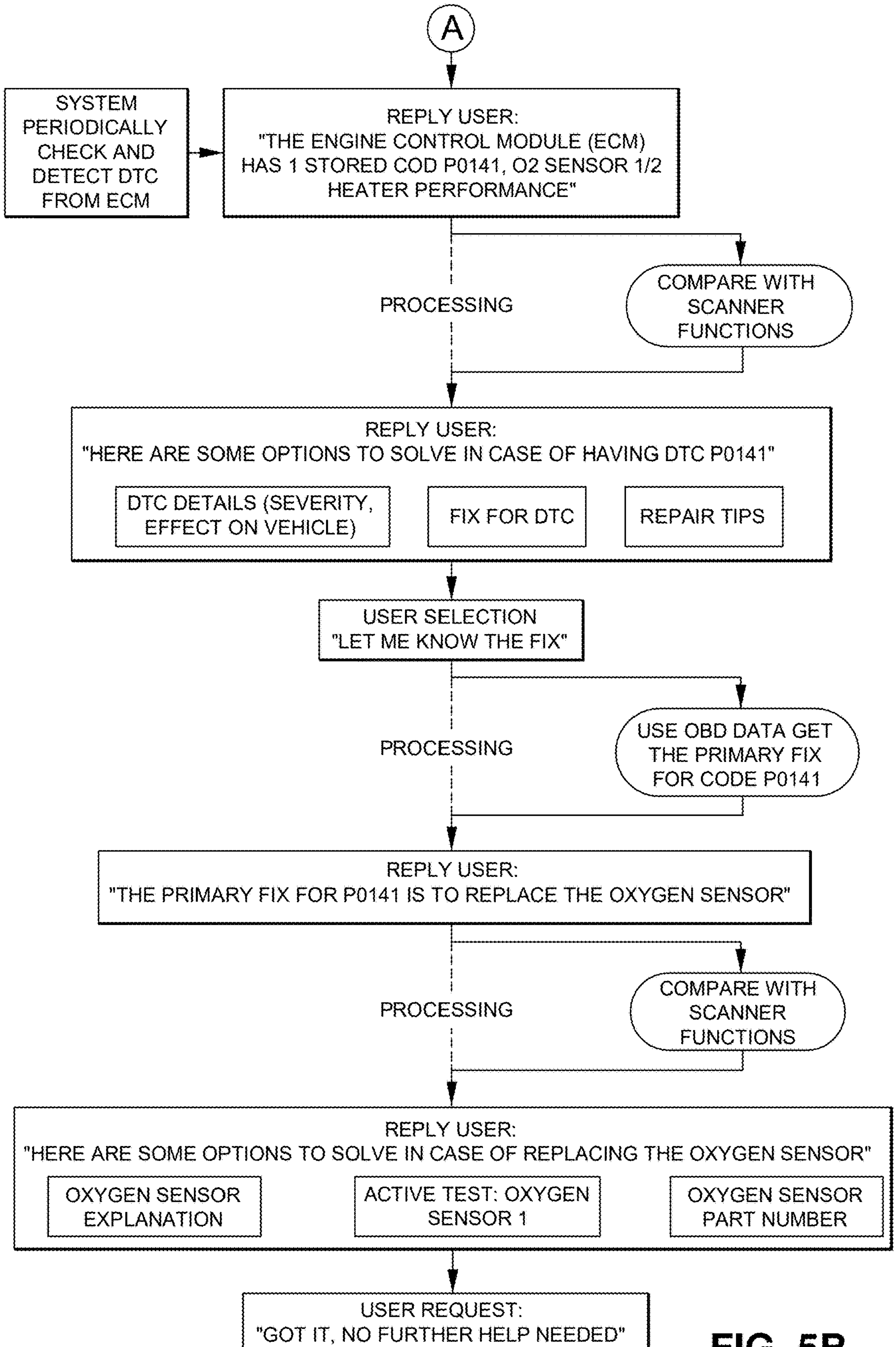

A
SYSTEM PERIODICALLY CHECK AND DETECT DTC FROM ECM
REPLY USER: "THE ENGINE CONTROL MODULE (ECM) HAS 1 STORED COD P0141, O2 SENSOR 1/2 HEATER PERFORMANCE"
PROCESSING
COMPARE WITH SCANNER FUNCTIONS
REPLY USER: "HERE ARE SOME OPTIONS TO SOLVE IN CASE OF HAVING DTC P0141"
DTC DETAILS (SEVERITY, EFFECT ON VEHICLE)
FIX FOR DTC
REPAIR TIPS
USER SELECTION "LET ME KNOW THE FIX"
PROCESSING
USE OBD DATA GET THE PRIMARY FIX FOR CODE P0141
REPLY USER: "THE PRIMARY FIX FOR P0141 IS TO REPLACE THE OXYGEN SENSOR"
PROCESSING
COMPARE WITH SCANNER FUNCTIONS
REPLY USER: "HERE ARE SOME OPTIONS TO SOLVE IN CASE OF REPLACING THE OXYGEN SENSOR"
OXYGEN SENSOR EXPLANATION
ACTIVE TEST: OXYGEN SENSOR 1
OXYGEN SENSOR PART NUMBER
USER REQUEST: "GOT IT, NO FURTHER HELP NEEDED"

FIG. 5B

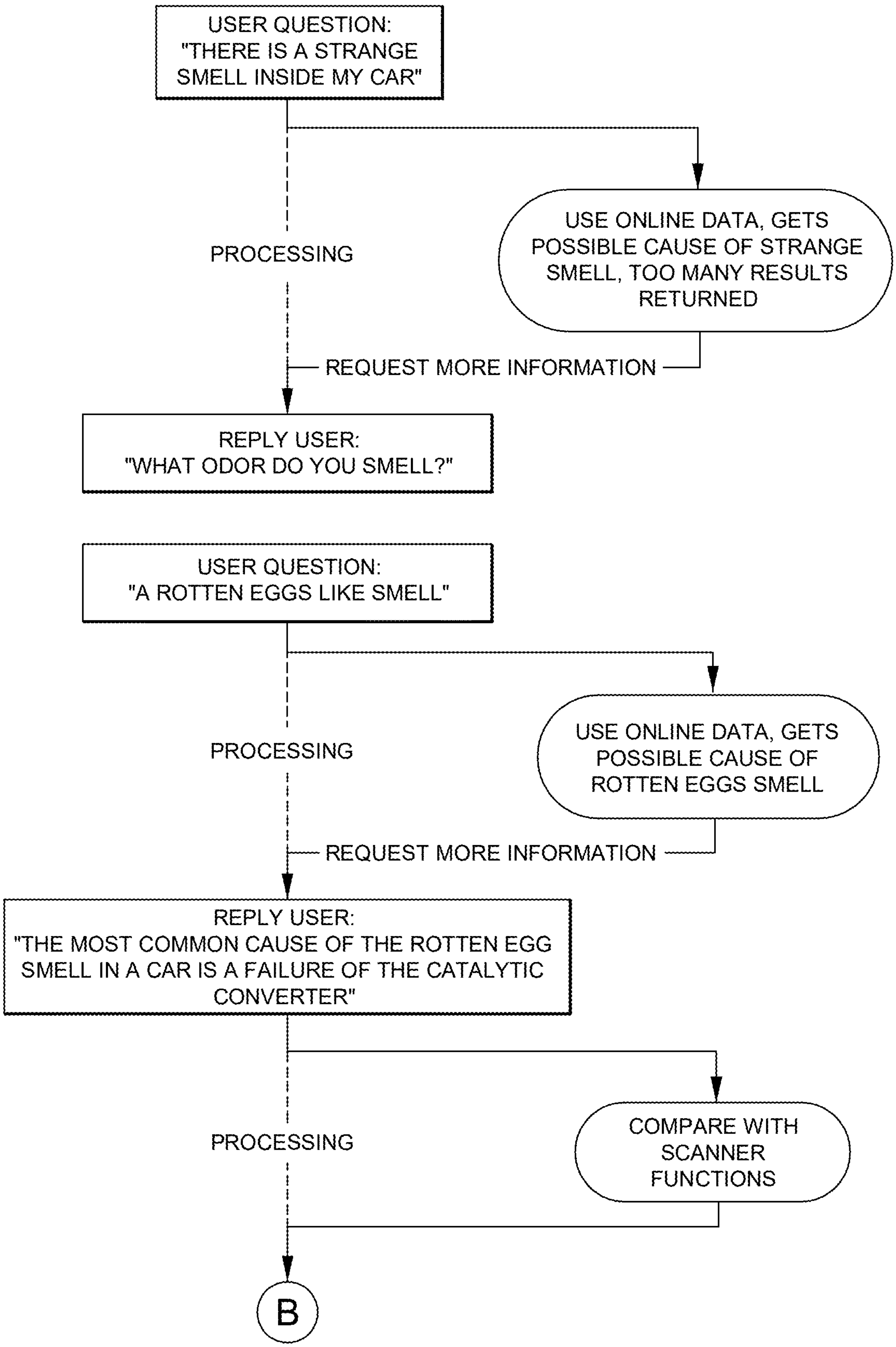

USER QUESTION:
"THERE IS A STRANGE SMELL INSIDE MY CAR"
PROCESSING
USE ONLINE DATA, GETS POSSIBLE CAUSE OF STRANGE SMELL, TOO MANY RESULTS RETURNED
REQUEST MORE INFORMATION
REPLY USER:
"WHAT ODOR DO YOU SMELL?"
USER QUESTION:
"A ROTTEN EGGS LIKE SMELL"
PROCESSING
USE ONLINE DATA, GETS POSSIBLE CAUSE OF ROTTEN EGGS SMELL
REQUEST MORE INFORMATION
REPLY USER:
"THE MOST COMMON CAUSE OF THE ROTTEN EGG SMELL IN A CAR IS A FAILURE OF THE CATALYTIC CONVERTER"
PROCESSING
COMPARE WITH SCANNER FUNCTIONS
B

FIG. 6A

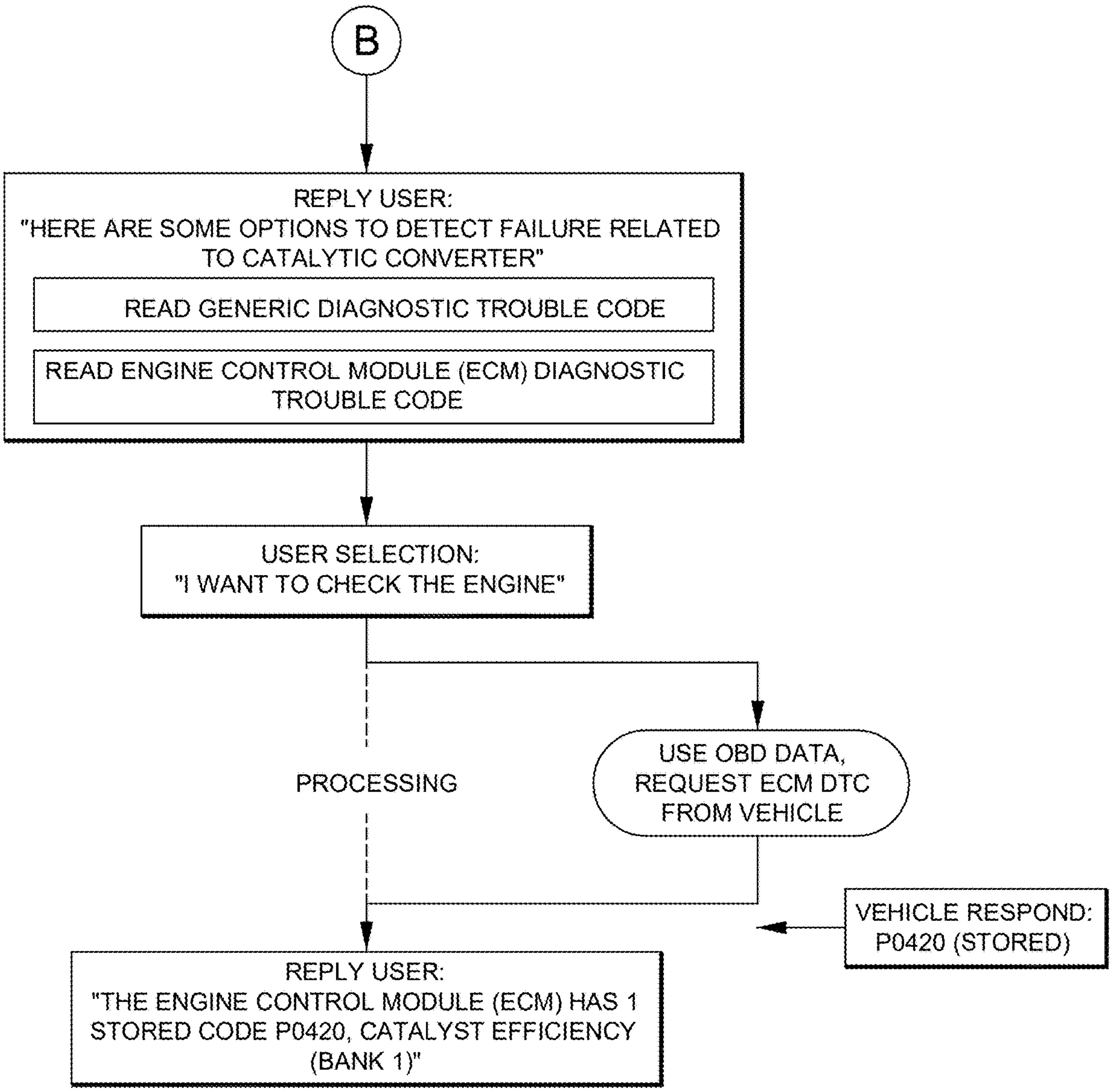

B
REPLY USER:
"HERE ARE SOME OPTIONS TO DETECT FAILURE RELATED TO CATALYTIC CONVERTER"
READ GENERIC DIAGNOSTIC TROUBLE CODE
READ ENGINE CONTROL MODULE (ECM) DIAGNOSTIC TROUBLE CODE
USER SELECTION:
"I WANT TO CHECK THE ENGINE"
PROCESSING
USE OBD DATA, REQUEST ECM DTC FROM VEHICLE
VEHICLE RESPOND: P0420 (STORED)
REPLY USER:
"THE ENGINE CONTROL MODULE (ECM) HAS 1 STORED CODE P0420, CATALYST EFFICIENCY (BANK 1)"

FIG. 6B

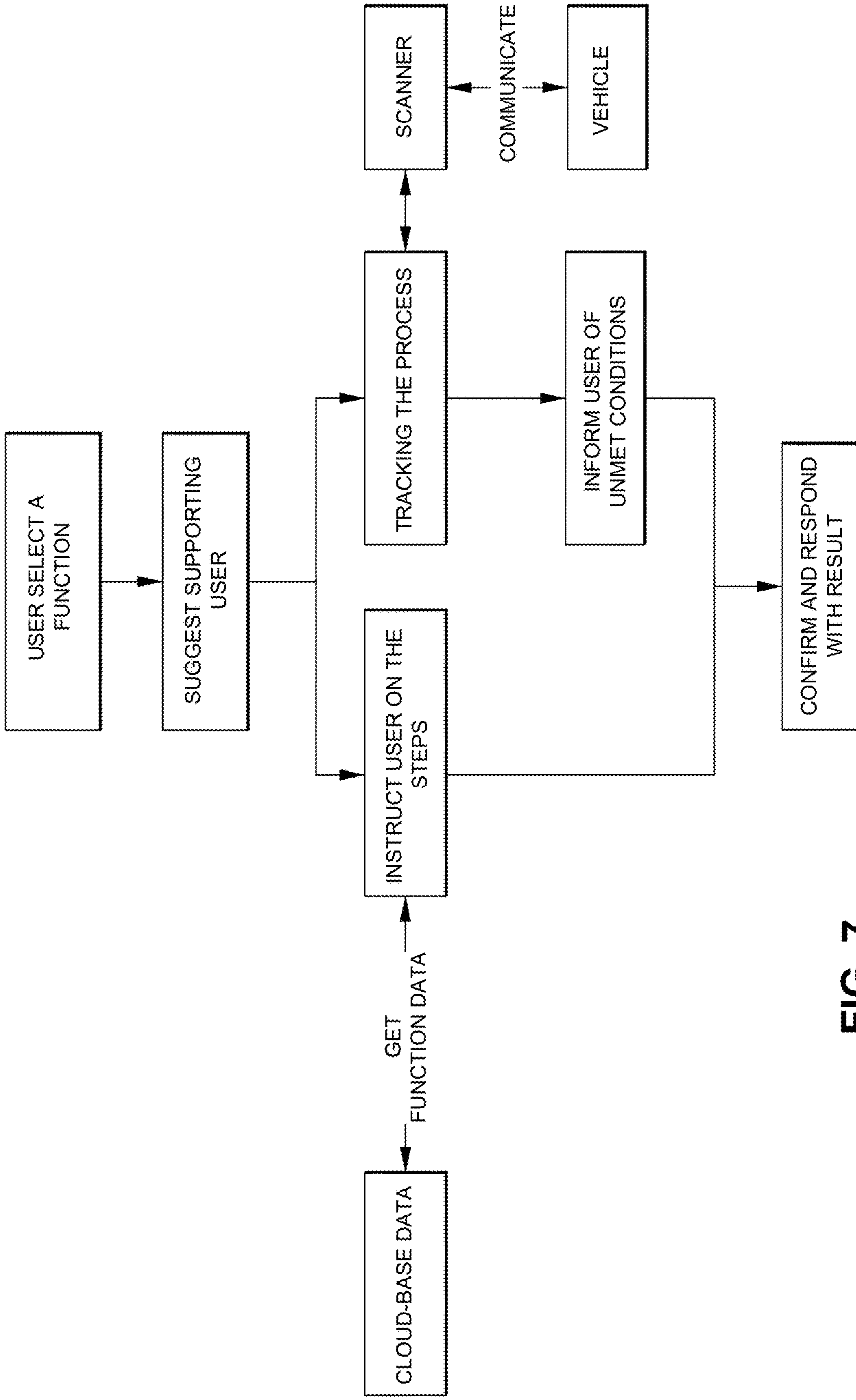

USER SELECT A FUNCTION
SUGGEST SUPPORTING USER
TRACKING THE PROCESS
SCANNER
COMMUNICATE
VEHICLE
INFORM USER OF UNMET CONDITIONS
INSTRUCT USER ON THE STEPS
GET FUNCTION DATA
CLOUD-BASE DATA
CONFIRM AND RESPOND WITH RESULT

FIG. 7

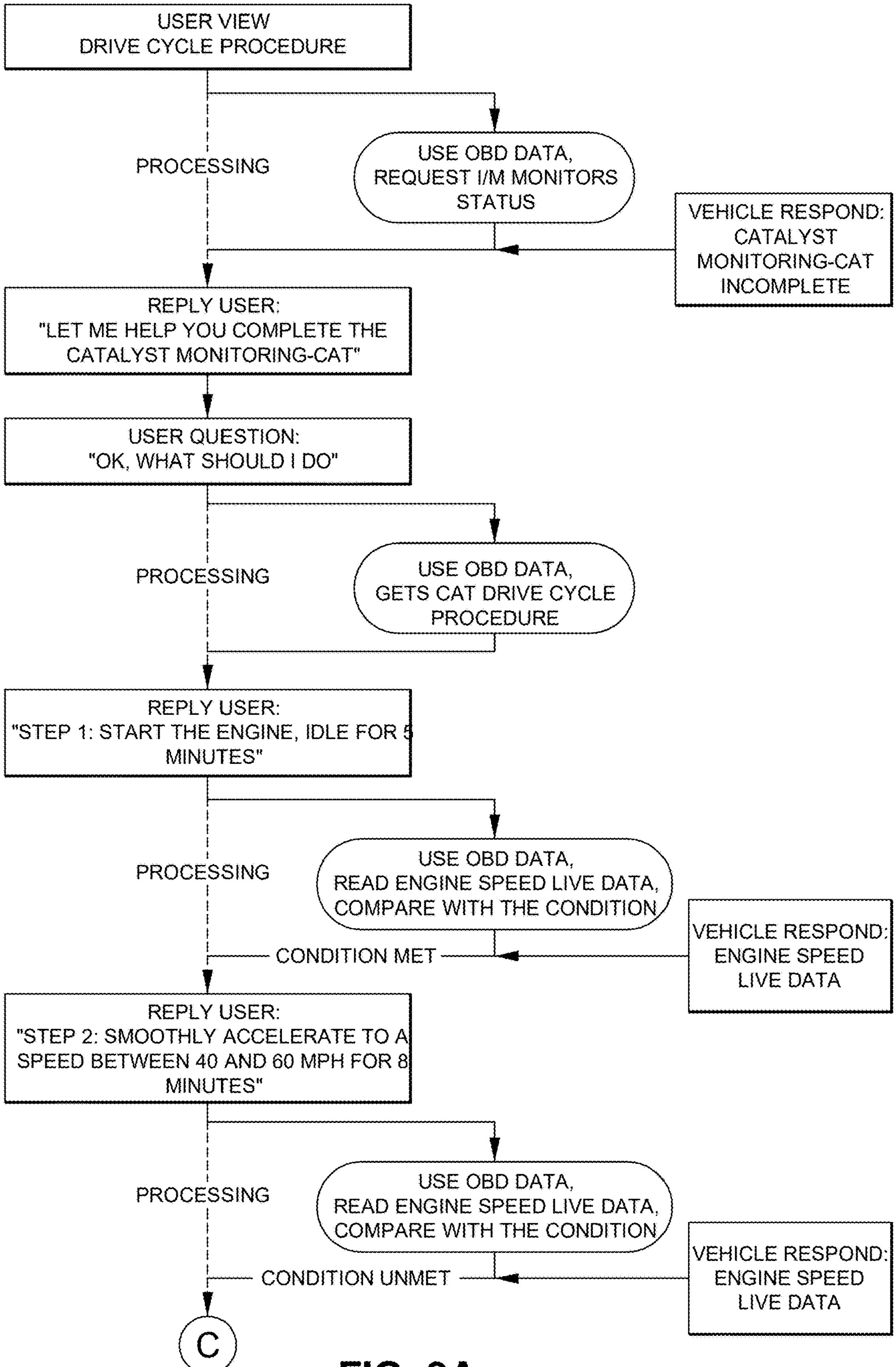

USER VIEW
DRIVE CYCLE PROCEDURE
PROCESSING
USE OBD DATA, REQUEST I/M MONITORS STATUS
VEHICLE RESPOND: CATALYST MONITORING-CAT INCOMPLETE
REPLY USER: "LET ME HELP YOU COMPLETE THE CATALYST MONITORING-CAT"
USER QUESTION: "OK, WHAT SHOULD I DO"
PROCESSING
USE OBD DATA, GETS CAT DRIVE CYCLE PROCEDURE
REPLY USER: "STEP 1: START THE ENGINE, IDLE FOR 5 MINUTES"
PROCESSING
USE OBD DATA, READ ENGINE SPEED LIVE DATA, COMPARE WITH THE CONDITION
CONDITION MET
VEHICLE RESPOND: ENGINE SPEED LIVE DATA
REPLY USER: "STEP 2: SMOOTHLY ACCELERATE TO A SPEED BETWEEN 40 AND 60 MPH FOR 8 MINUTES"
PROCESSING
USE OBD DATA, READ ENGINE SPEED LIVE DATA, COMPARE WITH THE CONDITION
CONDITION UNMET
VEHICLE RESPOND: ENGINE SPEED LIVE DATA
C

FIG. 8A

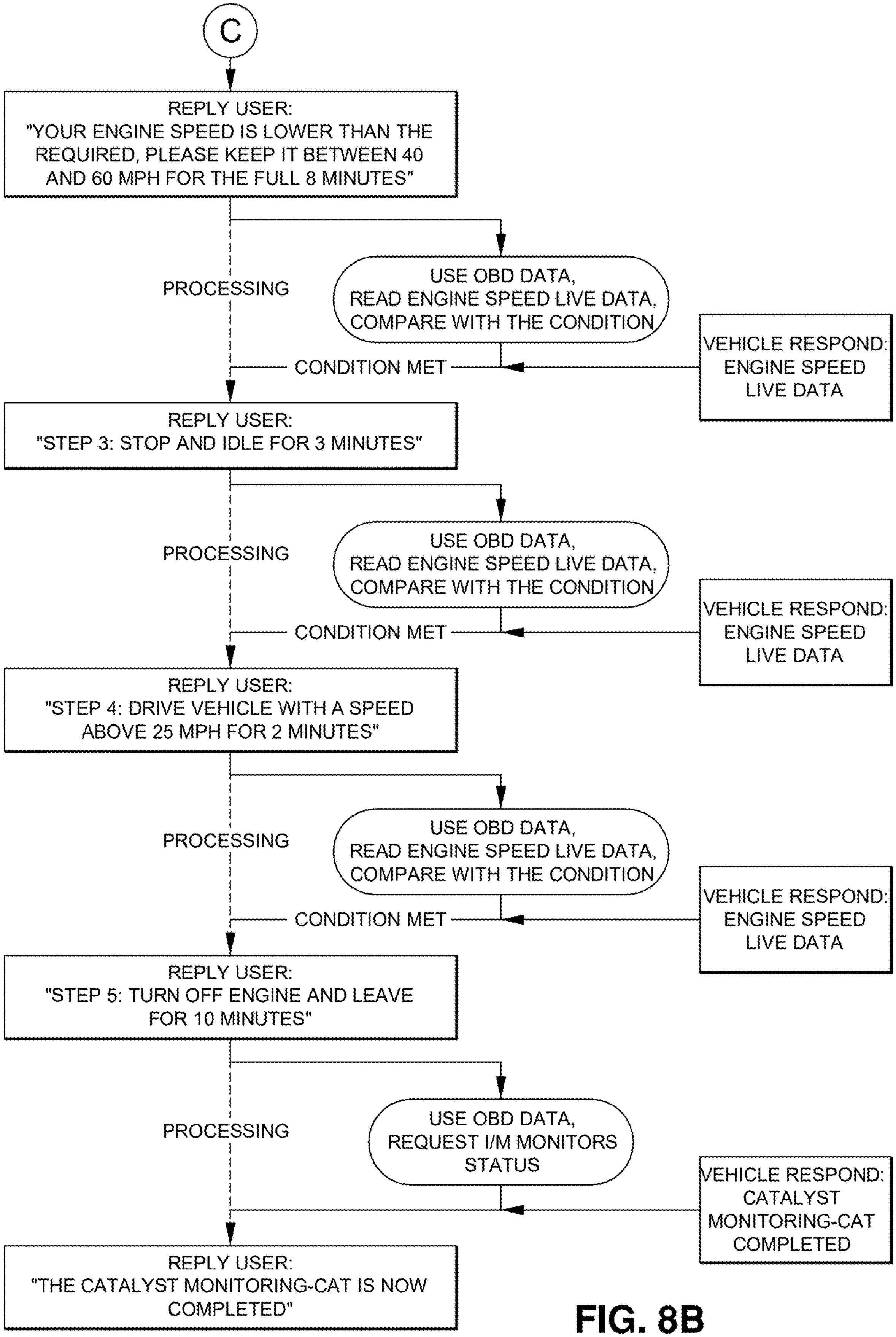

C
REPLY USER: "YOUR ENGINE SPEED IS LOWER THAN THE REQUIRED, PLEASE KEEP IT BETWEEN 40 AND 60 MPH FOR THE FULL 8 MINUTES"
PROCESSING
USE OBD DATA, READ ENGINE SPEED LIVE DATA, COMPARE WITH THE CONDITION
VEHICLE RESPOND: ENGINE SPEED LIVE DATA
CONDITION MET
REPLY USER: "STEP 3: STOP AND IDLE FOR 3 MINUTES"
PROCESSING
USE OBD DATA, READ ENGINE SPEED LIVE DATA, COMPARE WITH THE CONDITION
VEHICLE RESPOND: ENGINE SPEED LIVE DATA
CONDITION MET
REPLY USER: "STEP 4: DRIVE VEHICLE WITH A SPEED ABOVE 25 MPH FOR 2 MINUTES"
PROCESSING
USE OBD DATA, READ ENGINE SPEED LIVE DATA, COMPARE WITH THE CONDITION
VEHICLE RESPOND: ENGINE SPEED LIVE DATA
CONDITION MET
REPLY USER: "STEP 5: TURN OFF ENGINE AND LEAVE FOR 10 MINUTES"
PROCESSING
USE OBD DATA, REQUEST I/M MONITORS STATUS
VEHICLE RESPOND: CATALYST MONITORING-CAT COMPLETED
REPLY USER: "THE CATALYST MONITORING-CAT IS NOW COMPLETED"

FIG. 8B

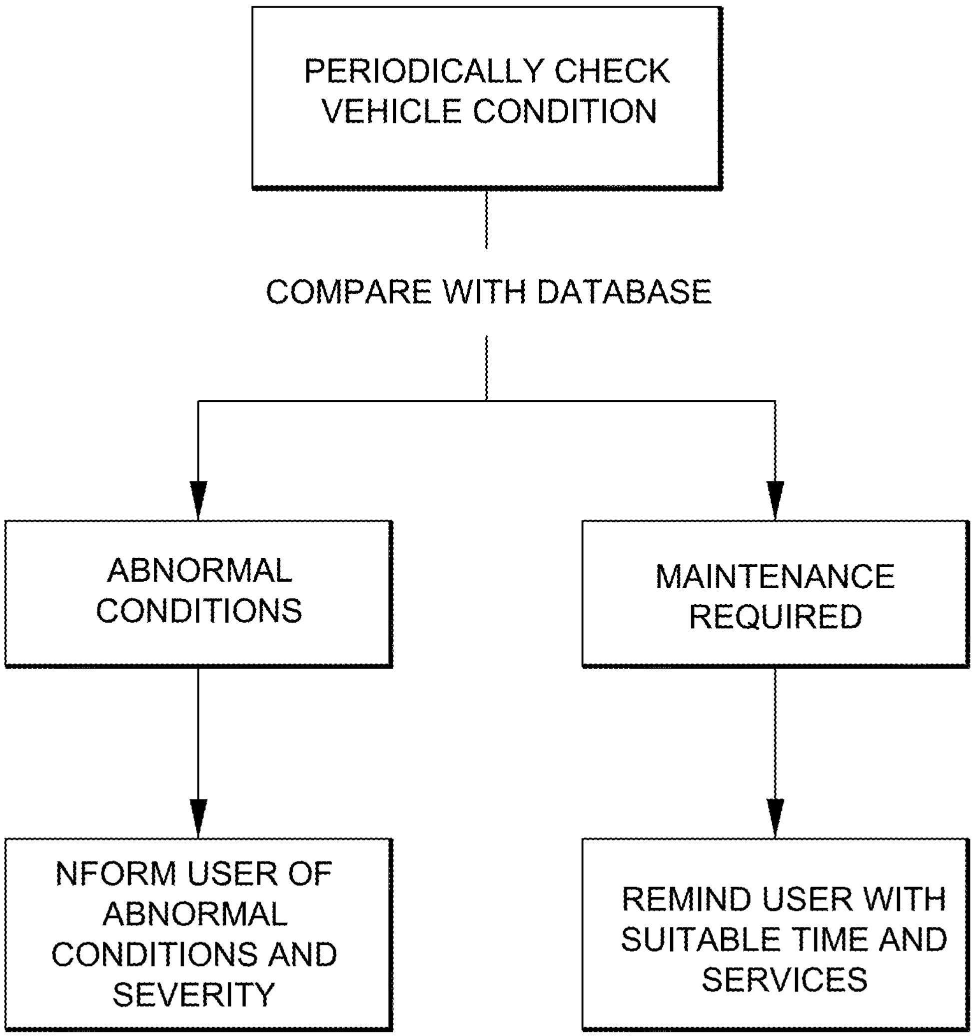

PERIODICALLY CHECK VEHICLE CONDITION
COMPARE WITH DATABASE
ABNORMAL CONDITIONS
MAINTENANCE REQUIRED
NFORM USER OF ABNORMAL CONDITIONS AND SEVERITY
REMIND USER WITH SUITABLE TIME AND SERVICES

FIG. 9

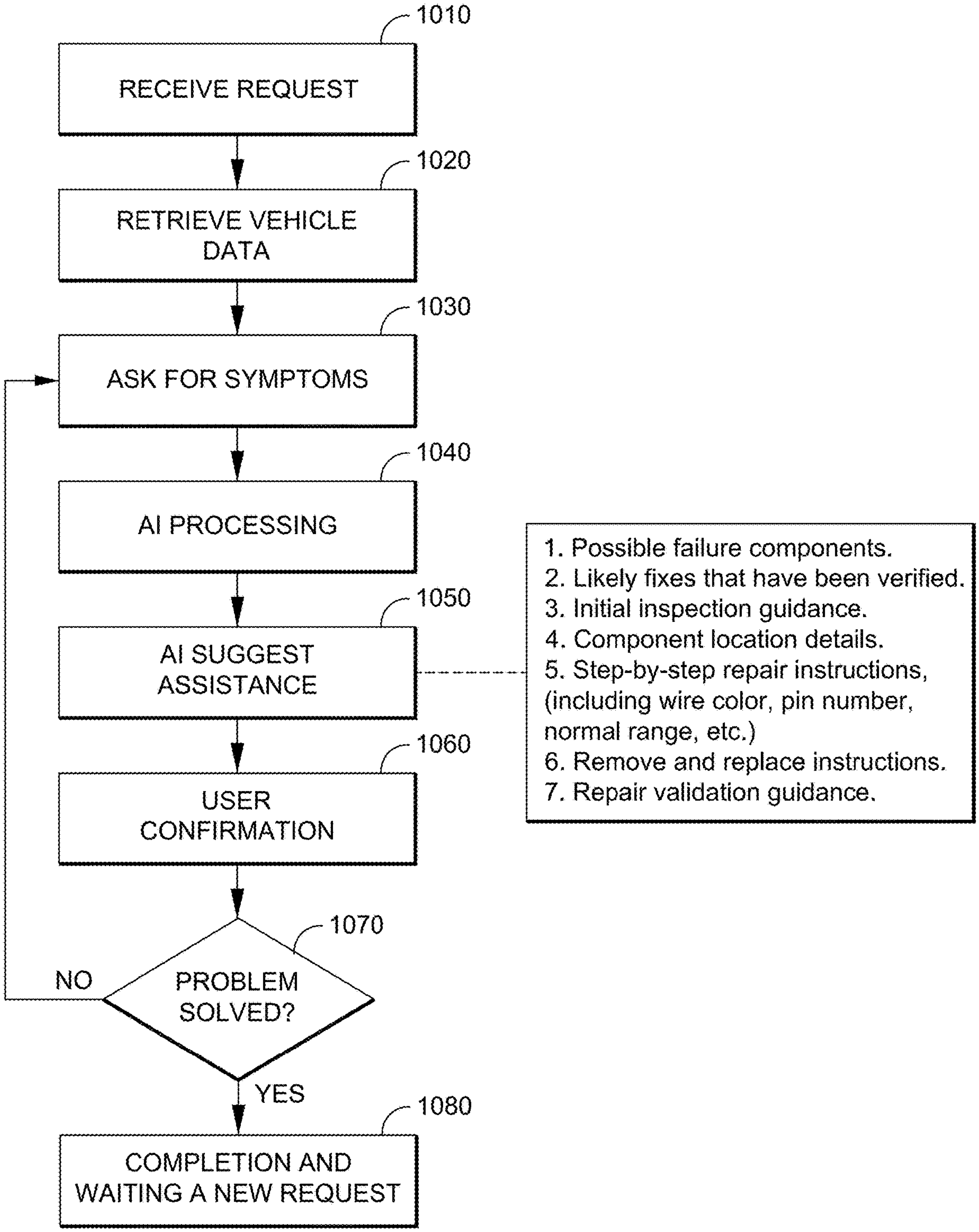
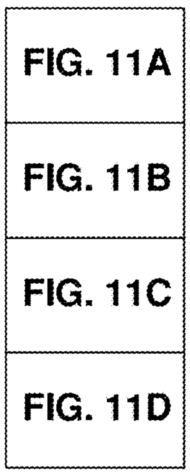

1010
RECEIVE REQUEST
1020
RETRIEVE VEHICLE DATA
1030
ASK FOR SYMPTOMS
1040
AI PROCESSING
1050
AI SUGGEST ASSISTANCE
1060
USER CONFIRMATION
1070
PROBLEM SOLVED?
NO
YES
1080
COMPLETION AND WAITING A NEW REQUEST
1. Possible failure components.
2. Likely fixes that have been verified.
3. Initial inspection guidance.
4. Component location details.
5. Step-by-step repair instructions, (including wire color, pin number, normal range, etc.)
6. Remove and replace instructions.
7. Repair validation guidance.

FIG. 10

FIG. 11



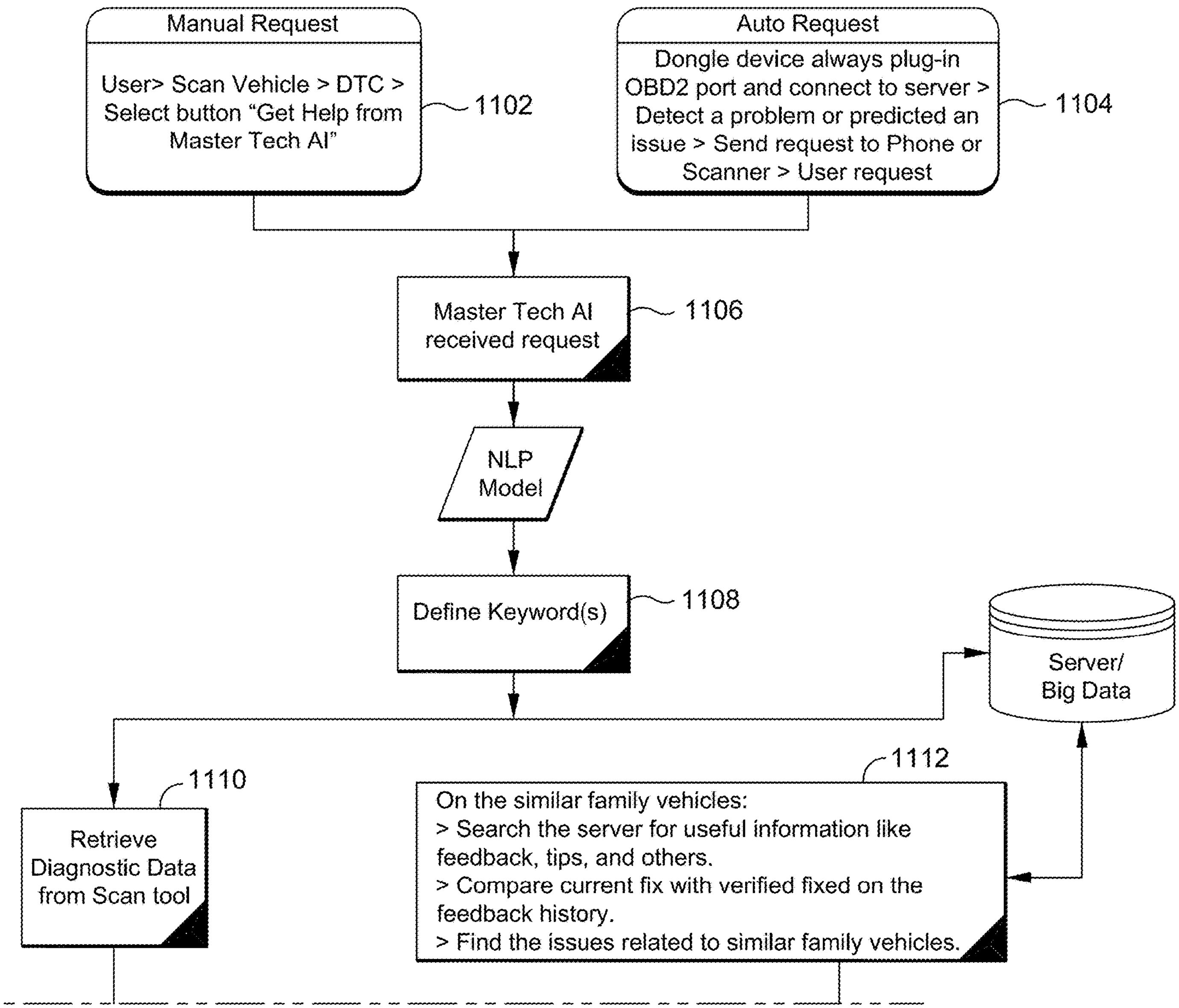
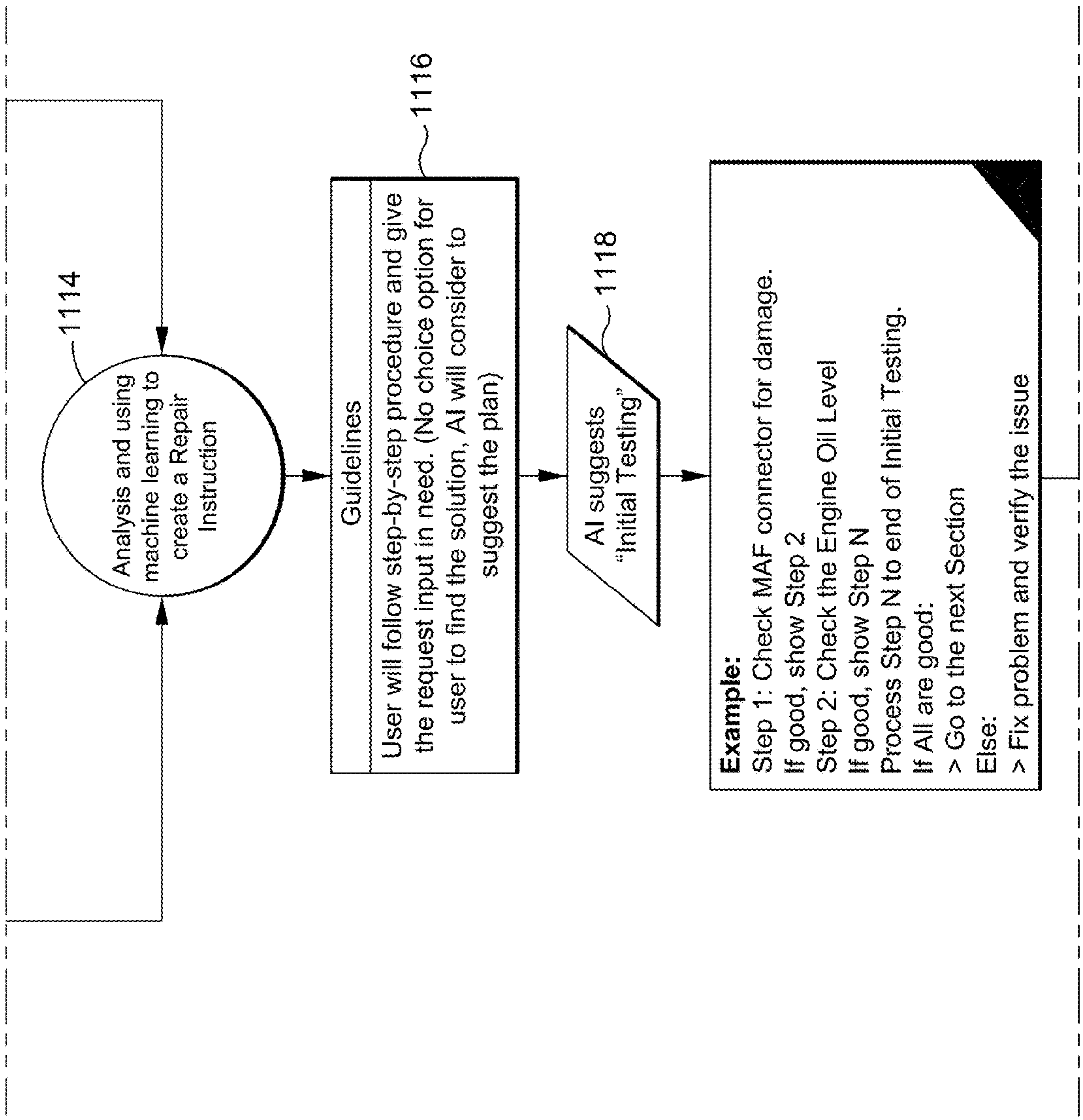


FIG. 11A

FIG. 11B


1114
Analysis and using machine learning to create a Repair Instruction
1116
Guidelines
User will follow step-by-step procedure and give the request input in need. (No choice option for user to find the solution, AI will consider to suggest the plan)
1118
AI suggests "Initial Testing"
Example:
Step 1: Check MAF connector for damage.
If good, show Step 2
Step 2: Check the Engine Oil Level
If good, show Step N
Process Step N to end of Initial Testing.
If All are good:
> Go to the next Section
Else:
> Fix problem and verify the issue

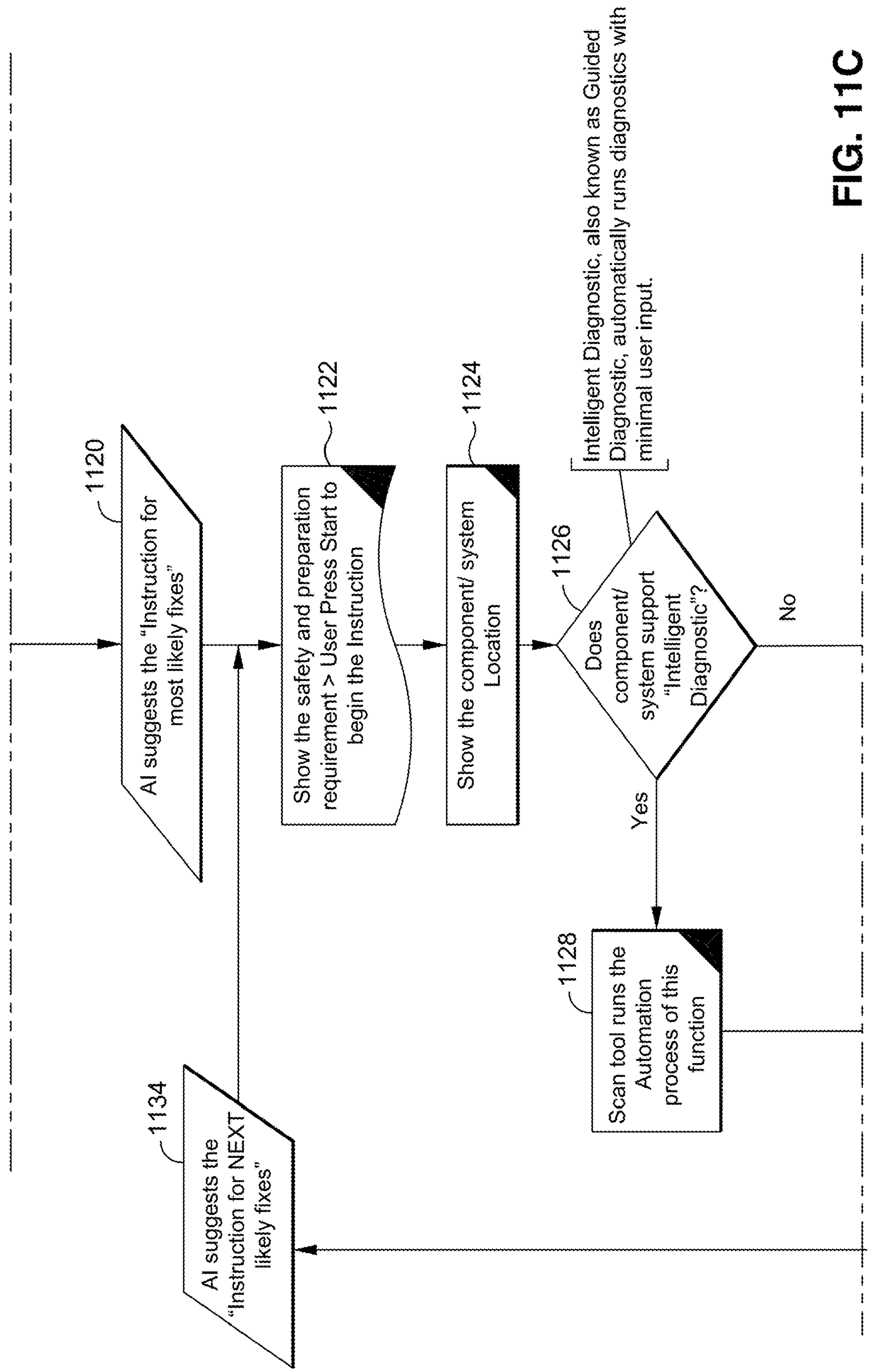

1120
AI suggests the "Instruction for most likely fixes"
1122
Show the safety and preparation requirement > User Press Start to begin the Instruction
1124
Show the component/ system Location
1126
Does component/ system support "Intelligent Diagnostic"?
Intelligent Diagnostic, also known as Guided Diagnostic, automatically runs diagnostics with minimal user input.
Yes
No
1128
Scan tool runs the Automation process of this function
1134
AI suggests the "Instruction for NEXT likely fixes"

FIG. 11C

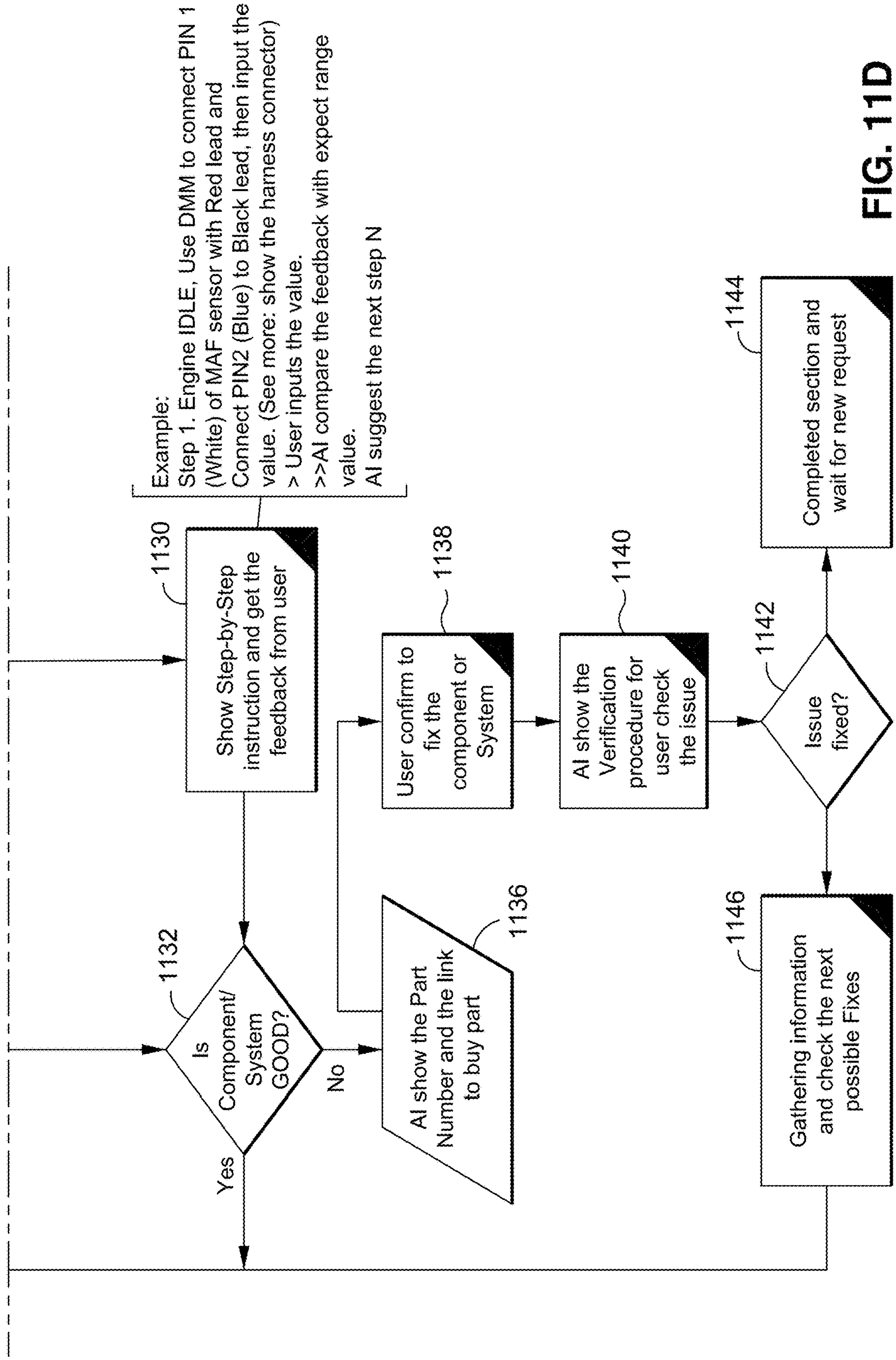

Example:
Step 1. Engine IDLE, Use DMM to connect PIN 1 (White) of MAF sensor with Red lead and Connect PIN2 (Blue) to Black lead, then input the value. (See more: show the harness connector)
> User inputs the value.
>>AI compare the feedback with expect range value.
AI suggest the next step N
1130
Show Step-by-Step instruction and get the feedback from user
1132
Is Component/ System GOOD?
Yes
No
1136
AI show the Part Number and the link to buy part
1138
User confirm to fix the component or System
1140
AI show the Verification procedure for user check the issue
1142
Issue fixed?
1144
Completed section and wait for new request
1146
Gathering information and check the next possible Fixes

FIG. 11D

ARTIFICIAL INTELLIGENCE ASSISTANT FOR VEHICLE DIAGNOSTICS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application claims priority to U.S. Provisional Application Ser. No. 63/558,537, filed Feb. 27, 2024, the entire contents of which is incorporated by reference herein.

STATEMENT RE: FEDERALLY SPONSORED RESEARCH/DEVELOPMENT

[0002] Not Applicable

BACKGROUND

[0003] Automotive scan tools have the potential to empower vehicle owners to take matters into their own hands when it comes to vehicle diagnostics. Using a scan tool, a vehicle owner can retrieve diagnostic data from the vehicle's various systems and access a diagnostic database to derive a diagnostic condition of the vehicle including, for example, a most likely root cause of a problem that the vehicle owner is experiencing. In this way, the vehicle owner can streamline the process of servicing the vehicle and/or finding the necessary parts to repair the vehicle. As automotive diagnostic technology continues to improve, the functionality of scan tools has expanded to include various kinds of scans and data retrieval functions. The more functionality that scan tools have, the more powerful they are, but the more difficult they are to understand and use, especially in the hands of vehicle owners who are not professional auto mechanics. As a result, the full benefits of scan tools are not realized as vehicle owners either fail to take advantage of the scan tool's increased functionality or skip the scan altogether and take the vehicle in to the shop at the first sign of trouble, even in cases where professional service might not have been necessary.

[0004] By the same token, existing scan tools that are available for use by the general public, such as those found at kiosks in auto parts stores, may generally be configured to have only a limited set of features for ease of use by non-professionals. As such, the customers of these stores are unable to take advantage of the more powerful features that would be useful to more efficiently and accurately diagnose their vehicles.

[0005] In addition, even when equipped with the diagnostic functionality of a scan tool, a user may be unable to accurately diagnose an issue occurring in the vehicle without combining the diagnostic data with a degree of human expertise. While it may be possible to search the Internet for additional information pertaining to a particular diagnostic trouble code (DTC) or other scan tool output, the information is generally non-specific to the user's vehicle, making it difficult and time-consuming to find the root cause of the issue.

BRIEF SUMMARY

[0006] The present disclosure contemplates various systems and methods for overcoming the above drawbacks accompanying the related art. A vehicle owner, technician, or other user of the disclosed subject matter may engage in ordinary conversation with the vehicle or with an application (e.g., a mobile app) associated with the vehicle that acts as a gateway connection between the user and the vehicle. It may be that the user is experiencing a problem with the vehicle or currently witnessing an unusual noise or other symptom of a possible vehicle defect. Or it may be that the user would like assistance with the vehicle or simply wants information about the vehicle. By processing the user's spoken (or typed) commands, questions, and concerns using a natural language processing (NLP) model, the disclosed app is able to derive appropriate scan tool feature functions to efficiently address the situation, without the user needing to know how to operate a scan tool or other data acquisition and transfer device (DAT). Even the most advanced scan tool functions and special purpose features may be automatically selected and/or recommended and explained to the user for confirmation as appropriate. The user can benefit from the full range of capabilities found in professional scan tools without any level of expertise or even a working knowledge of correct vehicle terminology. By employing NLP to expand upon the user's utterances and introduce related technical terminology, the system is able to derive keywords or other inputs for determining the most relevant scan tool feature functions and, ultimately, the diagnostic condition of the vehicle including the root cause of a problem, repair solutions, and/or replacement parts.

[0007] One aspect of the embodiments of the present disclosure is a computer program product comprising one or more non-transitory program storage media on which are stored instructions executable by one or more processors or programmable circuits to perform operations for providing vehicle diagnostics. The operations may comprise receiving a first communication from a user, processing the first communication using a natural language processing (NLP) model to produce NLP model output including a sequence of words, extracting one or more keywords from the sequence of words, and instructing a data acquisition and transfer device (DAT) connected to an OBD port of a vehicle to perform a selected function based on the one or more keywords.

[0008] Another aspect of the embodiments of the present disclosure is a method of providing vehicle diagnostics. The method may comprise receiving a first communication from a user, processing the first communication using an NLP model to produce NLP model output including a sequence of words, extracting one or more keywords from the sequence of words, and instructing a DAT connected to an OBD port of a vehicle to perform a selected function based on the one or more keywords.

[0009] Another aspect of the embodiments of the present disclosure is a system for providing vehicle diagnostics. The system may comprise a mobile device operable to receive a first communication from a user, process the first communication using an NLP model to produce NLP model output including a sequence of words, and extract one or more keywords from the sequence of words. The system may further comprise a DAT in communication with the mobile device and connected to an OBD port of a vehicle. The DAT may be operable to perform a selected function based on the one or more keywords.

[0010] Another aspect of the embodiments of the present disclosure is a system for providing vehicle diagnostics. The system may comprise one or more servers operable to receive a first communication from a user, process the first communication using a natural language processing (NLP) model to produce NLP model output including a sequence of words, and extract one or more keywords from the sequence of words. The system may further comprise a DAT in communication with the one or more servers and connected to an OBD port of a vehicle. The DAT may be operable to perform a selected function based on the one or more keywords.

[0011] The operations and/or methods of any of the above aspects of the embodiments described herein may comprise receiving diagnostic data from the DAT, the diagnostic data having been retrieved from the vehicle by the DAT in accordance with the selected function, and determining a diagnostic condition of the vehicle based on the diagnostic data. Determining the diagnostic condition may include uploading the diagnostic data to one or more servers and receiving the diagnostic condition from the one or more servers. The diagnostic condition may be derived based on a historical database of vehicle-specific diagnostic data and associated diagnostic conditions and/or using artificial intelligence (AI). The operations and/or methods may comprise deriving a reliability index associated with the diagnostic condition. The operations and/or methods may comprise, in response to the reliability index being below a threshold, instructing the DAT to perform a special function test or presenting to the user an option to instruct the DAT to perform a special function test. The special function test may be configured to generate additional diagnostic data that supports or refutes the determined diagnostic condition. The diagnostic condition of the vehicle may comprise an indication of a repair solution and/or an indication of a replacement part.

[0012] Processing the first communication using the NLP model may include supplementing the first communication with a context related to the vehicle. The context may include identifying information of the vehicle, past user communications, state information of the vehicle, and/or environmental information that may be captured from various sensors including accelerometers and other sensors found in typical smartphones and tablets.

[0013] The operations and/or methods may comprise determining one or more proposed functions based on the one or more keywords, receiving a second communication from a user, and processing the second communication to determine the selected function from among the one or more proposed functions. Determining the one or more proposed functions may include mapping the one or more keywords to the one or more proposed functions. Determining the one or more proposed functions may include inputting the one or more keywords to a machine learning model whose output indicates the one or more proposed functions (e.g., using artificial intelligence). The one or more proposed functions may include a special function test. The operations and/or methods may comprise presenting a presumed diagnostic condition to the user. The special function test may be configured to generate additional diagnostic data that supports or refutes the presumed diagnostic condition.

[0014] The operations and/or methods may comprise determining the selected function by mapping the one or more keywords to the selected function.

[0015] The operations and/or methods may comprise determining the selected function by inputting the one or more keywords to a machine learning model whose output indicates the selected function.

[0016] Another aspect of the embodiments of the present disclosure is a computer program product comprising one or more non-transitory program storage media on which are stored instructions executable by one or more processors or programmable circuits to perform operations for providing vehicle diagnostics. The operations may comprise receiving a request for AI-assisted diagnostics, receiving diagnostic data obtained from a vehicle, prompting a user to describe one or more symptoms exhibited by the vehicle, processing the one or more symptoms and the diagnostic data at least in part using a natural language processing (NLP) model, and providing natural language guidance to the user based on a result of the processing.

[0017] The diagnostic data may be obtained from the vehicle in response to the request being received. The request may be derived from speech of the user and/or text entered by the user. The diagnostic data may be obtained autonomously in response to the request being received. The diagnostic data may be obtained from the vehicle at predefined intervals, and the request may be generated in response to an evaluation of the diagnostic data. The request may be generated by a scan tool. The diagnostic data may include identification information of the vehicle (e.g., a VIN). The one or more symptoms may be derived from speech of the user and/or text entered by the user. The natural language guidance may include testing instructions for the user to perform a step-by-step procedure to diagnose the vehicle. The testing instructions may include a wire color, a pin number, and/or an expected range value. The testing instructions may include prompting the user to input a measurement value, and the operations may further comprise comparing the measurement value to an expected range value. The operations may further comprise initiating an automated diagnostic routine based on the result of the processing. The natural language guidance may be provided to the user further based on a result of the automated diagnostic routine. The natural language guidance may further include repair instructions for the user to perform a repair on the vehicle. The repair instructions may include identification information of a replacement part. The repair instructions may include a verification procedure for confirming that the repair was successful. The processing and the providing of the natural language guidance may proceed autonomously in response to receipt of the one or more symptoms. The processing of the one or more symptoms and the diagnostic data may include uploading the one or more symptoms and the diagnostic data to one or more servers and receiving the result from the one or more servers. The processing may include inputting the one or more symptoms and the diagnostic data to a machine learning model. The processing may include generating instructions to collect additional data based on an output of the machine learning model.

[0018] Another aspect of the embodiments of the present disclosure is a system comprising the above computer program product and a mobile device. The mobile device may include the one or more processors or programmable circuits. The diagnostic data may be obtained from the vehicle by a dongle communicatively coupled with the mobile device.

[0019] Another aspect of the embodiments of the present disclosure is a system comprising the above computer program product and a scan tool. The scan tool may include the one or more processors or programmable circuits and may be operable to obtain the diagnostic data from the vehicle.

[0020] Another aspect of the embodiments of the present disclosure is a method of providing vehicle diagnostics. The method may comprise receiving a request for AI-assisted diagnostics, receiving diagnostic data obtained from a vehicle, prompting a user to describe one or more symptoms exhibited by the vehicle, processing the one or more symptoms and the diagnostic data at least in part using a natural language processing (NLP) model, and providing natural language guidance to the user based on a result of the processing.

[0021] Another aspect of the embodiments of the present disclosure is a system for providing vehicle diagnostics. The system may comprise a mobile device operable to receive a request for AI-assisted diagnostics, receive diagnostic data obtained from a vehicle, and prompt a user to describe one or more symptoms exhibited by the vehicle. The system may further comprise one or more servers in communication with the mobile device. The one or more servers may be operable to process the one or more symptoms and the diagnostic data at least in part using a natural language processing (NLP) model. The mobile device may be further operable to provide natural language guidance to the user based on a result of the processing.

[0022] The system may comprise a data acquisition and transfer device (DAT) in communication with the mobile device and connected to an OBD port of the vehicle. The DAT may be operable to obtain the diagnostic data from the vehicle.

[0023] Another aspect of the embodiments of the present disclosure is a system for providing vehicle diagnostics. The system may comprise a scan tool operable to receive a request for AI-assisted diagnostics, obtain diagnostic data from a vehicle, and prompt a user to describe one or more symptoms exhibited by the vehicle. The system may further comprise one or more servers in communication with the scan tool. The one or more servers may be operable to process the one or more symptoms and the diagnostic data at least in part using a natural language processing (NLP) model. The scan tool may be further operable to provide natural language guidance to the user based on a result of the processing.

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] These and other features and advantages of the various embodiments disclosed herein will be better understood with respect to the following description and drawings, in which like numbers refer to like parts throughout, and in which:

[0025] FIG. **1** shows a system for providing vehicle diagnostics according to one or more embodiments of the present disclosure;

[0026] FIG. **2** shows an operational flow for providing vehicle diagnostics according to one or more embodiments of the present disclosure;

[0027] FIG. **3** shows a sub-operational flow of step **240** in FIG. **2**;

[0028] FIG. **4** shows a flow chart describing an example application of the system of FIG. **1** and/or the operational flow of FIG. **2**;

[0029] FIG. **5**A shows a first part of a flow chart describing another example application of the system of FIG. **1** and/or the operational flow of FIG. **2**;

[0030] FIG. **5**B shows a second part of the flow chart of FIG. **5**A;

[0031] FIG. **6**A shows a first part of another flow chart describing an example application of the system of FIG. **1** and/or the operational flow of FIG. **2**;

[0032] FIG. **6**B shows a second part of the flow chart of FIG. **6**A;

[0033] FIG. **7** shows another flow chart describing an example application of the system of FIG. **1** and/or the operational flow of FIG. **2**;

[0034] FIG. **8**A shows a first part of another flow chart describing an example application of the system of FIG. **1** and/or the operational flow of FIG. **2**;

[0035] FIG. **8**B shows a second part of the flow chart of FIG. **8**A;

[0036] FIG. **9** shows another flow chart describing an example application of the system of FIG. **1** and/or the operational flow of FIG. **2**;

[0037] FIG. **10** shows an operational flow of an automotive artificial intelligence (AI) assistant system according to one or more embodiments of the present disclosure;

[0038] FIG. **11** shows an overview of a detailed operational flow corresponding to the operational flow of FIG. **10**;

[0039] FIG. **11**A shows a first portion of the detailed operational flow;

[0040] FIG. **11**B shows a second portion of the detailed operational flow;

[0041] FIG. **11**C shows a third portion of the detailed operational flow; and

[0042] FIG. **11**D shows a fourth portion of the detailed operational flow.

DETAILED DESCRIPTION

[0043] The present disclosure encompasses various embodiments of systems and methods for providing vehicle diagnostics using an intelligent communication interface. The detailed description set forth below in connection with the appended drawings is intended as a description of several currently contemplated embodiments and is not intended to represent the only form in which the disclosed invention may be developed or utilized. The description sets forth the functions and features in connection with the illustrated embodiments. It is to be understood, however, that the same or equivalent functions may be accomplished by different embodiments that are also intended to be encompassed within the scope of the present disclosure. It is further understood that the use of relational terms such as first and second and the like are used solely to distinguish one from another entity without necessarily requiring or implying any actual such relationship or order between such entities.

[0044] FIG. **1** shows a system **100** for providing vehicle diagnostics according to one or more embodiments of the present disclosure. The system **100** may allow a vehicle owner with little or no automotive technical knowledge or experience with scan tool feature functionality to easily locate and initiate the most effective scan tool functions quickly and efficiently when diagnosing the vehicle **10**. Referring also to the operational flow of FIG. **2**, a diagnostic methodology using the system **100** may begin with receiving a first communication **12** from the vehicle owner or other user (step **210**). For example, in a case where the diagnostic methodology is performed by a mobile device **110** such as a smartphone or tablet having an installed application (e.g., a mobile app), the application may enable the mobile device **110** to receive the user's communication **12** via a microphone thereof (in the case of a spoken communication **12**) or via a wireless communication interface thereof (in the case of a communication **12** in the form of a text message). While the communication modality (e.g., speech, typed text, etc.) may depend on the particular scenario, it is contemplated that the communication **12** may advantageously comprise natural language, such that the user may be able to converse freely with the mobile device **110** as opposed to being constrained to specific words and phrases that are preprogrammed into the application. Under the control of the app, the mobile device **110** may then, as described in more detail below, initiate one or more functions of a data acquisition and transfer device (DAT) **120** that may be connected to an on-board diagnostics (OBD) port of the vehicle **10**, e.g., a data link connector (DLC) port. The DAT **120** may be a dongle that plugs into the OBD port or a scanner or scan tool that connects to the OBD port via a cable, for example, to collect diagnostic data from an onboard computer (e.g., an electronic control unit or ECU) of the vehicle **10** for diagnostic analysis. The mobile device **110** may communicate with the DAT **120** using a shortrange wireless communication protocol (e.g., Bluetooth) or a wired connection, for example.

[0045] Given that the user may be inexperienced with automotive technology and scan tool functionality, the first communication **12** uttered by the user may not be well formulated. It may, for example, be vague, incomplete, or ambiguous and/or may use unconventional (or incorrect) terminology. Example communications **12** may include "The car won't start" or "What's going on with my brakes?" or may provide even less context such as "What's that clicking sound?" or "What does this light mean?" Even given the capability of parsing these questions into words (e.g., by an automatic speech recognition or speech to text algorithm), these communications **12** lack necessary information that would, in principle, be needed to meaningfully instruct a DAT **120**. To this end, after the first communication **12** is received from the user, the diagnostic methodology of FIG. **2** may proceed with processing the communication **12** using a natural language processing (NLP) model **130** to produce NLP model output (step **220**). In a readily deployable embodiment of the disclosed subject matter, it is envisioned that a pretrained machine learning model such as a general purpose chatbot or virtual assistant (e.g., OpenAI's ChatGPT) may serve as the NLP model **130**. The user's communication **12** may be uploaded or otherwise provided by the mobile device **110** as input to the NLP model **130**, and the NLP model **130** may create NLP model output including a sequence of words, typically in the form of a human-readable response to the communication **12** that attempts to answer the user's question or otherwise address the user's need. As can be appreciated, such NLP model output may to varying degrees (depending on the particular NLP model **130**) be thorough, detailed, lengthy, and/or simulate natural language, but may not necessarily be reliably accurate. This may be understood to be due to the nature of natural language processing, which may be thought of as a tool for mimicking responses to similar questions without truly "knowing" the answer to the question posed. In response to the user's communication **12** of "The car won't start," for example, the NLP model output may be something like, "Here is a list of possible reasons that a vehicle won't start: the key fob has low battery or is not in range; the vehicle is low on fuel; the vehicle battery is depleted or faulty and may need to be replaced; there is a problem with the alternator; there is a problem with the timing belt; . . . " etc. As described in more detail below, the system **100** may make use of the NLP model output not for its conventional purpose of providing an answer or solution for the user but, rather, as a means of elaborating upon and finetuning the user's original communication **12** for use in deriving one or more suitable scan tool functions to perform.

[0046] In some cases, the step of processing the user's communication **12** using the NLP model **130** may include supplementing the communication **12** with a context related to the vehicle **10**. Given that the NLP model **130** may be a general-purpose model that is not designed specifically for dialogue about vehicles, it may be especially advantageous for the system **100** to provide a context for questions that would otherwise not be recognizable as having to do with a vehicle, such as "What's that clicking sound?" or "What does this light mean?" In a relatively simple embodiment, for example, the mobile device **110** (e.g., under the control of the app) may simply append a phrase such as "in a vehicle" or "for vehicle diagnostics" to the user's communication **12**. Thus, "What's that clicking sound?" or "What does this light mean?" may become "What's that clicking sound in a vehicle?" or "What does this light mean in a vehicle?" A context like this, which may have universal applicability for any envisioned communication **12** of the user, may be redundant in some cases (e.g., "The car won't start" becomes "The car won't start in a vehicle") but nondetrimental, while providing important information to the NLP model **130** in many cases where the context would otherwise not be clear. More specific contexts are also contemplated, such as where the context includes identifying information of the particular vehicle **10** in question. Identifying information of the vehicle **10** may include, for example, year/make/model/engine/trim information, which may be stored in a user profile associated with the app (e.g., locally or on a server), derived from a VIN retrieved from the vehicle **10** by the DAT **120**, or, in some cases, deduced by the system **100** from other diagnostic data of the vehicle **10**. By providing identifying information of the particular vehicle **10** to the NLP model **130** together with the user's communication **12**, it may be appreciated that the resulting NLP model output may be more relevant. Other types of context that may be used to supplement the user's communication **12** are also contemplated, such as a previous communication **12** of the user related to the vehicle **10** or a state (e.g., ON/OFF, vehicle speed, gear) of the vehicle **10**, which may be derived from diagnostic data retrieved from the vehicle **10** by the DAT **120**, for example. In some cases, a context may include environmental data derived from one or more sensors (e.g., a vapor sensor) that may be present in or near the vehicle **10** including sensors or other components of the mobile device **110**. For example, an accelerometer built into the mobile device **110** may indicate that acceleration is not smooth or that the vehicle, potentially pointing to a fuel line function or that the vehicle **10** is experiencing a bumpy ride, suggesting a tire pressure issue.

[0047] As noted above, the system **100** may make use of the NLP model output as a means of elaborating upon and finetuning the user's original communication **12**. In this regard, the operational flow of FIG. **2** may continue with extracting one or more keywords from the sequence of words included in the NLP model output (step **230**). For example, in the above scenario where the user said, "The car won't start," it was envisioned that the NLP model output might be something like, "Here is a list of possible reasons that a vehicle won't start: the key fob has low battery or is not in range; the vehicle is low on fuel; the vehicle battery is depleted or faulty and may need to be replaced; there is a problem with the alternator; there is a problem with the timing belt; . . . " etc. In this instance, the mobile device **110** might extract, for example, "key fob," "low," "battery," "range," "fuel," "depleted," "alternator," "timing belt" (note that the word pairs "key fob" and "timing belt" may be tokenized as key words using a tokenization algorithm that recognizes them as two-word terms based on the statistical occurrences of the words), and/or might similarly extract keywords that include phrases such as "vehicle won't start," "low battery," "low on fuel," "battery is depleted," etc. The resulting list of keywords may represent a brainstorm of symptoms and potential diagnostic conditions that are closely related to the user's original communication **12**. In this way, the NLP model output may be thought of not only as a response to the user's communication **12** (which may be the conventional purpose of the NLP model output as used in other contexts) but also, from the perspective of the system **100**, as a detailed reformulation of the underlying concern that prompted the user's communication **12** or, in other words, as a rephrasing of the user's request. That is, the system **100** may be less interested in the articulated response output by the NLP model **130** and more interested in mining the NLP model output for raw associations (in the form of keywords) to expand on and develop the user's original communication **12**. The sequence of words output by the NLP model **130** may thus be repurposed as input to a computer-implemented vehicle diagnostic methodology rather than presented to a human user as may be the original purpose of third-party NLP model output, for example.

[0048] With the keyword(s) having been extracted from the NLP model output, the operational flow of FIG. **2** may continue with instructing the DAT **120** based on the keyword (s) (step **240**). The mobile device **110** (e.g., under the control of the app) may, for example, communicate with the DAT **120** via a wired or wireless connection to instruct the DAT **120** to perform a selected function based on the keyword(s). The available functions may depend on the manufacturer, with functions being added or optimized from time to time, and may include, by way of example, any or all of the following scan tool functions or variants thereof: read and clear OBD fault codes, read freeze frame, read and clear original equipment manufacturer (OEM) fault codes, read OBD live data, read OEM live data, monitor status, drive cycle procedure, service check (e.g., warning light status, brake pad check, battery status, oil life status, transmission temperature, etc.), tire pressure monitoring system (TPMS) pressure and status, active test, special function, workshop tools, etc. Selection of scan tool functions based on the keyword(s) may be done by mapping the keyword(s) to the functions. For example, a set of possible functions indexed by keywords (e.g., in the form of a lookup table or decision tree) may be stored locally on the mobile device **110** and managed by the app or may be stored remotely and accessed by the app through communication with one or more servers **140** (e.g., over the Internet via a cellular or WiFi network, for example). By using the keyword(s) as an index, the app may select one or more functions. For example, the keywords "battery," "low battery," "vehicle won't start," "battery is depleted," etc. may indicate that the DAT **120** should perform a function to check battery status, while the keywords "check engine," "check engine light," etc. may indicate that the DAT **120** should perform function(s) to read one or more types of diagnostic data from the vehicle **10**. Additional example keywords might include "ABS code," "read tire pressure," "test battery," "check brake pad," "SRS light," "ignition," which may be matched or otherwise mapped to related scan tool functions. Instead of or in addition to using the keywords as an index, the keywords may be input to a machine learning model whose output indicates the one or more function(s), with the machine learning model having been trained on sets of keywords (and possibly associated diagnostic data) to generate a relevant set of function(s). In this way, artificial intelligence (AI) may be used to associate various keywords or combinations of keywords (which may be representative of vehicle-related concepts) with relevant scan tool functions. Environmental and other context data as described above, such as contemporaneously captured sensor data of the mobile device **110**, may also be input to the machine learning model to improve the relevance of the determined function(s).

[0049] To further resolve ambiguity and/or allow the user to participate in the diagnostic procedure, it is contemplated that the system **100** (e.g., the app) may propose one or more scan tool functions to the user for selection or confirmation. In this regard, FIG. **3** shows an example sub-operational flow of step **240**, which may begin with determining one or more proposed functions based on the one or more keywords (step **242**). The proposed functions may be presented to the user by the system **100** (e.g., via the app), on screen or preferably by verbal communication with the user (e.g., using a speaker of the mobile device **110**). In addition to the names of the proposed functions, the system **100** may provide the user with an explanation of each proposed function so that the user can make an informed choice. The sub-operational flow of FIG. **3** may continue with receiving a second communication **12** from the user, which may indicate a selection and/or confirmation of one or more of the proposed functions. Like the first communication **12** that initiated the process, the second communication **12** may be speech, typed text, etc. The second communication **12** may be processed by the system **100** to determine a selected function (step **246**), which may be used for instructing the scan tool **120** in step **240** of FIG. **2**. In a relatively simple embodiment, the second communication **12** may include a selection from an enumerated menu of options (e.g., by making a selection on screen or repeating a specific word or phrase in the case of spoken communication with the app), such that there is little or no possibility of misunderstanding the user's intent. The processing of the second communication **12** may in this case be straightforward (i.e., selecting the scan tool function that is unambiguously identified/confirmed by the user), though it may still involve an automatic speech recognition or speech to text algorithm, for example. Alternatively, natural language processing may be used to interpret the second communication **12**, allowing the user more flexibility with his/her wording. A flow chart describing an example application of the operational flow of FIG. **2** including the sub-operational flow of FIG. **3** is shown in FIG. **4**, in which the user confirms one of three proposed functions ("Function A," "Function B," and "Function C") for instructing the DAT **120**.

[0050] By virtue of the above methodology (with or without the user selection/confirmation described in relation to FIG. 3), the system 100 may allow a user with little or no understanding of scan tool functionality to effectively make use of modern scan tool functions, even as they may be continually expanded and updated by the manufacturer of the DAT 120. The user may simply talk to (or text with or otherwise communicate with) the app, which, from the user's perspective, may feel like the user is talking to the vehicle 10 as if it were a person. Owing to the use of the NLP model 120 to expand upon the user's vague or ambiguous utterances as described above, it may be entirely unnecessary for the user to know specific commands, and no particular level of detail or sophistication may be needed in the user's communication 12. The user may have the impression that the system 100 understands what the user means, no matter how imprecise the user's concern or request may be.

[0051] Once the DAT 120 has been instructed in step 240 of FIG. 2 and one or more scan tool functions have been initiated by the DAT 120 (see FIG. 1), the results of the function(s) may be used to diagnose the vehicle 10. In this regard, it may be appreciated that the DAT 120 or the mobile device 110 may, in general, upload diagnostic data collected from the vehicle 10 to the one or more servers 140, which may derive a diagnostic condition of the vehicle 10 from the uploaded diagnostic data by comparing the uploaded diagnostic data with data (e.g., historical data) stored in one or more diagnostic databases 150. The diagnostic condition of the vehicle 10 may include, for example, information about the root cause of a problem that the vehicle 10 is exhibiting and/or an indication of one or more repair solutions or replacement parts for addressing the problem. Exemplary diagnostic methods, including the use of such diagnostic data to arrive at a most likely root cause and repair solution as well as vehicle-specific replacement parts, are described in the following U.S. patent documents, each of which is owned by Innova Electronics Corporation of Irvine, California: U.S. Pat. No. 6,807,469, entitled AUTO DIAGNOSTIC METHOD AND DEVICE, U.S. Pat. No. 6,925,368, entitled AUTO DIAGNOSTIC METHOD AND DEVICE, U.S. Pat. No. 7,620,484, entitled AUTOMOTIVE MOBILE DIAGNOSTICS, U.S. Pat. No. 8,068,951, entitled VEHICLE DIAGNOSTIC SYSTEM, U.S. Pat. No. 8,019,503, entitled AUTOMOTIVE DIAGNOSTIC AND REMEDIAL PROCESS, U.S. Pat. No. 8,370,018, entitled AUTOMOTIVE DIAGNOSTIC PROCESS, U.S. Pat. No. 8,909,416, entitled HANDHELD SCAN TOOL WITH FIXED SOLUTION CAPABILITY, U.S. Pat. No. 9,014,908, entitled MULTI-STAGE DIAGNOSTIC SYSTEM AND METHOD, U.S. Pat. No. 9,142,066, entitled MULTI-STAGE DIAGNOSTIC SYSTEM AND METHOD, U.S. Pat. No. 9,026,400, entitled DIAGNOSTIC PROCESS FOR HOME ELECTRONIC DEVICES, U.S. Pat. No. 9,177,428, entitled PREDICTIVE DIAGNOSTIC METHOD, U.S. Pat. No. 9,646,432, entitled HAND HELD DATA RETRIEVAL DEVICE WITH FIXED SOLUTION CAPABILITY, U.S. Pat. No. 9,824,507, entitled MOBILE DEVICE BASED VEHICLE DIAGNOSTIC SYSTEM, U.S. Pat. No. 10,643,403, entitled PREDICTIVE DIAGNOSTIC METHOD AND SYSTEM, U.S. Pat. No. 11,068,560, entitled METHOD OF PROCESSING VEHICLE DIAGNOSTIC DATA, U.S. Pat. No. 11,270,529, entitled SYSTEM AND METHOD FOR PROACTIVE VEHICLE DIAGNOSIS AND OPERATIONAL ALERT, and U.S. Pat. No. 11,158,141, entitled SYSTEM AND METHOD FOR PROACTIVE VEHICLE DIAGNOSIS AND OPERATIONAL ALERT, the entire contents of each of which is expressly incorporated by reference herein. The operational flow of FIG. 2 may thus continue with receiving diagnostic data from the DAT 120 (step 250), the diagnostic data having been retrieved from the vehicle 10 by the DAT 120 in accordance with the selected function(s) of step 240, and determining a diagnostic condition of the vehicle 10 (step 260). Advantageously, the system 100 (e.g., the app) may determine the diagnostic condition based on the diagnostic data and, in some cases, further based on the one or more keywords that were extracted from the NLP model output in step 230. In this way, the diagnostic analysis performed by the system 100 may take into account both the expanded brainstorm of symptoms and potential diagnostic conditions derived by the NLP model 130 from the user's original communication 12 and the actual diagnostic data of the vehicle 10 collected by the DAT 120 for a more accurate and relevant diagnosis. Environmental and other context data as described above, such as contemporaneously captured sensor data of the mobile device 110, may also be taken into account to further improve the results. For example, diagnostic data might include airbag sensor data indicating that the airbag was inflated, which may be corroborated by accelerometer data of the mobile device 110 indicating that the vehicle 10 experienced a sudden change in velocity suggestive of a collision. By corroborating the diagnostic data with environmental data in this way, a situation may be avoided where the family members of a passenger are notified prematurely of a crash when really there was simply a faulty airbag sensor.

[0052] In lieu of, or in addition to the above-described techniques for deriving a diagnostic solution or diagnostic condition based on use of a vehicle specific historical database, artificial intelligence (AI) may be used to recognize associations between certain vehicle data (e.g., live data, patterns of live data and static data), NLP model output, and/or other vehicle operational and environmental conditions (e.g., sensed vibrations, sounds, vapors, temperatures, etc.) with the diagnostic solution or condition, thus enabling the determination of a diagnostic solution or condition without the need to reference the historical database. For example, a machine learning model may be trained using historical data such as the historical data stored in the diagnostic database(s) 150, such that the machine learning model becomes increasingly capable of associating patterns of input data with diagnostic conditions. Subsequently, for a given set of new input data, the system 100 may derive the diagnostic condition(s) of the vehicle 10 using the machine learning model without there being any need to further consult the historical data. The AI may further be operative to identify external resources suitable to further inform the user with respect to the diagnostic solution and establish communication with such resources via an appropriate and available communication pathway, such as a cellphone enabled pathway or a V2X pathway to V2X service providers and associated resources. That functionality may proceed autonomously in response to receipt of the vehicle data, in response to an evaluation of the urgency of the vehicle diagnostic solution, or on-demand, in response to a user input. Another exemplary application of the AI enabled diagnostic process would be to support and facilitate the evolution of advanced driver-assistance systems (ADAS), in relation to installation, testing and/or monitoring of those systems during driving conditions.

[0053] It is noted that all or a portion of the operational flow of FIG. **2** may be performed by the mobile device **110** with installed app as described above, in which case the mobile device **110** may, for example, process the first communication using the NLP model in step **220** by communicating with third-party computer(s) where the NLP model **130** might in some implementations be located. By the same token, the mobile device **110** may, for example, determine scan tool function(s) in step **242** or determine the diagnostic condition of the vehicle **10** in step **260** by communicating with one or more servers **140** to request needed information and/or analysis with reference to one or more databases **150**, to the extent that such resources are not stored locally on the mobile device **110**. In some implementations, the server(s) **140** may associate the vehicle **10** with the user of the mobile application through the use of a user profile linked to the VIN of the vehicle **10** as described in the following U.S. patent documents, each of which is owned by Innova Electronics Corporation of Irvine, California: U.S. Pat. No. 11,335,139, entitled SYSTEM AND METHOD FOR SELECTIVE VEHICLE DATA RETRIEVAL, U.S. Pat. No. 11,455,841, entitled SYSTEM AND METHOD FOR SELECTIVE VEHICLE DATA RETRIEVAL, and U.S. Patent Application Pub. No. 2023/0063381, entitled SYSTEM, METHOD, AND COMPUTER PROGRAM PRODUCT FOR PROVIDING APPLICATION-BASED ASSISTANCE WITH VEHICLE EMISSION TEST COMPLIANCE, the entire contents of each of which is expressly incorporated by reference herein. Just as all or a portion of the operational flow of FIG. **2** may be performed by the mobile device **110**, it is contemplated that all or a portion of the operational flow of FIG. **2** may be performed by the one or more servers **140**. In this regard, the server(s) **140** may communicate with the DAT **120** and/or access the NLP model **130** directly or via the app installed on the mobile device **110**.

[0054] All or portions of the above-described operational flow of FIG. **2** may be performed iteratively to enable a back-and-forth conversation between the user and the system **100**. An extended example showing one such iterative application of the disclosed subject matter is shown in the flow chart of FIGS. **5**A and **5**B. The example may begin with the user detecting that there is a light illuminated on the dashboard and asking what is happening with the dashboard light. From the user's perspective, the system **100** (e.g., the app) may reply verbally to the user, "Your Check Engine light is ON, which means a potential problem with the engine or emissions system," followed by "Here are some options to detect where the problem is: Read Generic Diagnostic Trouble Code (DTC), Read Engine Control Module (ECM) Diagnostic Trouble Code (DTC), Scan All Systems," at which point the system **100** may wait for further input from the user. Behind the scenes, the processing steps may involve one or more iterations of the operational flow described in relation to FIG. **2** or portions thereof. To process the initial question about the dashboard light, the mobile device **110** with the installed app may initially use OBD data to determine which dashboard light is currently lit by requesting a warning light status using the DAT **120**. For example, the initial communication **12** "What happen with the dashboard light?" may be processed using the NLP model **130** to derive keywords, and the keywords may be mapped to the scan tool function of requesting the warning light status. Alternatively, in some implementations the app may first check whether the initial question itself has suitable keywords (e.g., "dashboard light") for selecting a suitable scan tool function, in which case the NLP model **130** may not be needed for this initial step. Upon receiving the OBD data from the vehicle **10** indicating that the check engine light is ON, the app may perform further processing to provide additional information to the user and come up with a list of relevant scan tool functions as shown in FIG. **4**A. Advantageously, the system **100** may derive this explanatory information and proposed scan tool functions by supplementing the user's initial communication **12** with the actual vehicle data indicating that the check engine light is on, processing the resulting query using the NLP model **130**, deriving keyword(s) from the NLP model output, and communicating with one or more server(s) **140** to retrieve the information and proposed functions from one or more diagnostic databases **150**.

[0055] In response to a second communication **12** from the user selecting one of the proposed scan tool functions, "I want to check the engine" (which may entail a simple selection from an enumerated menu of options or involve further natural language processing as described above), the system **100** (e.g., the app) may initiate the scan tool function of "request ECM DTC" from the vehicle **10** as shown in FIG. **4**A, after which the app may reply to the user to provide the result of the function as shown in FIG. **4**B, "The Engine Control Module (ECM) has 1 stored code P0141, O2 Sensor 1/2 Heater Performance." To arrive at this result, the app may again communicate with the one or more server(s)/database(s) **140**, **150** to upload the DTCs (along with vehicle identifying information derived from the VIN) and determine an associated diagnostic condition of the vehicle **10** (e.g., for the specific year/make/model/engine/trim). The app may again present the user with further options to pursue when having DTC P0141, some or all of which may involve additional scan tool functions as may be derived from the combined diagnostic data, user communication(s) **12**, and/or keyword(s) extracted from NLP model output. The user may then select from these options ("DTC Details (Severity, Effect on Vehicle), Fix for DTC, Repair Tips") as shown ("Let me know the fix"), and the system **100** may retrieve the fix from the server(s)/database(s) **140**, **150** and announce or otherwise present it to the user ("The Primary Fix for P0141 is to replace the oxygen sensor").

[0056] In this way, the system **100** may determine the diagnostic condition of the vehicle **10** in accordance with step **260** of FIG. **2** and present the diagnostic condition to the user, verbally and/or on screen on the mobile device **110**. Depending on the available functions of the DAT **120**, the user may continue to drill down and perform additional diagnostic functions as may be recommended by the system **100** and selected or confirmed by the user. As shown, for example, the flow chart of FIG. **4**B continues with the presentation to the user of options for further processing (which may be retrieved from the server(s)/database(s) **140**, **150** as described above): "Here are some options to solve in case of replacing the oxygen sensor: Oxygen Sensor Explanation, Active Test: Oxygen Sensor1, Oxygen Sensor Part Number." The user may, for example, choose to conduct a special function test such as "Active Test: Oxygen Sensor1," which would result in the system **100** retrieving more specific diagnostic data from the vehicle **10** using the DAT

120, which would then be used to retrieve a more specific diagnostic condition of the vehicle **10** for presentation to the user. The more specific diagnostic data may include, for example, live data, which may be found to support or refute a presumed diagnostic condition (such as a faulty oxygen sensor). In this way, an iterative application of the disclosed subject matter may enable a conversational guided diagnostic procedure, with the user able to drill down as desired to delve further into potential root causes through a back-and-forth dialogue with the system **100**. In the example of FIG. **4**B, the user instead chooses to exit the conversation by saying "Got it, no further help needed."

[0057] As noted above, the user may be presented with the option of performing a special function test as part of a dialogue with the system **100**. For example, a special function test may be performed that is associated with a vehicle component that is relevant to the diagnostic condition determined or presumed by the system **100**. In this regard, the diagnostic condition or solution derived by the system **100** (e.g., by operation of the DAT **120**, user communications **12**, and/or other data sources) can be confirmed by guided diagnostic testing, in a manner as set forth in U.S. patent application Ser. No. 18/328,289 ("the '289 application"), entitled SYSTEM AND METHOD FOR GUIDED VEHICLE DIAGNOSTICS, filed on Jun. 2, 2023 and owned by Innova Electronics Corporation of Irvine, California, the entire contents of which is expressly incorporated by reference. It is contemplated that the operations performed by the system **100** (e.g., by the app and/or server(s) **140**) may include deriving a reliability index associated with the presumed diagnostic condition of the vehicle. The reliability index may be a score (e.g., a numerical score) that quantifies a confidence in the diagnostic condition (or any other result) that is determined by the system **100** and may be based on, for example, historical data indicative of how often the same diagnostic condition (or other result) was found to be accurate in the same or similar vehicles under the same or similar circumstances in the past. In response to the reliability index being below a threshold, the system **100** (e.g., the app) may instruct the DAT **120** to perform a special function test (e.g., autonomously) or may present the user with an option to instruct the DAT **120** to perform a special function test, with the special function test being configured to generate additional diagnostic data that supports or refutes the determined diagnostic condition (e.g., by the collection of live data). A low reliability index may cause the system **100** to present options for performing a special function test, whereas a high reliability index (e.g., above some threshold) may cause the system **100** to bypass that option and proceed under the assumption that the presumed diagnostic condition is correct. Depending on the specific implementation of the system **100** and/or user preferences, autonomous performance of the special function test may proceed directly from the initial user communication or other event that resulted in the presumed diagnosis (e.g., the collection of diagnostic data from the vehicle **10** and/or other corroborating data as described above), independent of further human interaction. In the case of a special function test yielding diagnostic data that refutes a presumed diagnostic condition, the procedure may repeat with a second-most likely diagnostic condition being the new presumed diagnostic condition, followed by a third-most likely diagnostic condition, and so on, with additional special function tests being performed as needed or desired.

[0058] As another example, a reliability index may be assigned to the system's determination of a proposed scan tool function to perform as described above. A low reliability index may cause the system **100** to present one or more functions for user confirmation (as in the example operational flow of FIG. **3** and as illustrated in the flow charts of FIGS. **4** and **5**A/**5**B), whereas a high reliability index (e.g., above some threshold) may cause the system **100** to bypass the step of choosing/confirming the function and proceed with instructing the DAT **120** based on a single determined function having high confidence. It is noted, however, that some functions may still require confirmation for safety reasons, even in the case of a high reliability index. For example, some special functional tests (e.g., write-function type) need to be performed under certain conditions such as while the vehicle is stopped. The system **100** may be configured to warn the user of such conditions or any associated effects or dangers rather than bypassing user confirmation in some instances.

[0059] Another extended example showing an iterative application of the disclosed subject matter is shown in the flow chart of FIGS. **6**A and **6**B. In this case, the user begins by saying, "There's a strange smell inside my car." Upon processing this initial communication **12** using the NLP model **130**, the system **100** may determine that there are too many possible causes for a strange smell and that more information from the user is needed. This may be determined, for example, during step **230** of the operational flow of FIG. **2**, where it may be found that too many different keywords associated with different scan tool functions are found in the NLP model output (e.g., the number of extracted keywords or the number of distinct mapped functions is greater than some threshold). Therefore, the system **100** (e.g., the app) may reply verbally to the user, "What odor do you smell?" to get more specific information from the user. The user responds, "A rotten eggs like smell," which is processed using the NLP model **130**. The system **100** may, for example, process this new communication **12** using the NLP model **130** while also providing the previous communication **12** ("There's a strange smell inside my car") to the NLP model **130** as a context, such that the combination ("There's a strange smell inside my car. A rotten eggs like smell") may be processed by the NLP model **130** to derive NLP model output. Based on the NLP model output (e.g., based on keywords derived therefrom), the system **100** may inform the user, "The most common cause of the rotten egg smell in a car is a failure of the catalytic converter," and, as shown in FIG. **6**B, may propose scan tool functions ("Here are some options to detect failure related to catalytic converter: Read Generic Diagnostic Trouble Code (DTC), Read Engine Control Module (ECM) Diagnostic Trouble Code (DTC)"). In response to a second communication **12** from the user selecting one of the proposed scan tool functions, "I want to check the engine" (which may entail a simple selection from an enumerated menu of options or involve further natural language processing as described above), the system **100** (e.g., the app) may initiate the scan tool function of "request ECM DTC" from the vehicle **10**, after which the app may reply to the user to provide the result of the function as shown in FIG. **6**B, "The Engine Control Module (ECM) has 1 stored code P0420, Catalyst Efficiency (Bank 1)." To arrive at this result, the mobile device **110** with the installed app may instruct the DAT **120** to request ECM DTC from the vehicle **10** and may communicate with the one or more server(s)/database(s) **140**, **150** to upload the DTCs (along with vehicle identifying information derived from the VIN) and translate the DTCs and/or determine an associated diagnostic condition of the vehicle **10** (e.g., for the specific year/make/model/engine/trim).

[0060] FIG. **7** shows another flow chart describing an example application of the system **100** of FIG. **1** and/or the operational flow of FIG. **2**. When the user selects a scan tool function (or, alternatively, when the system **100** autonomously determines the function), the system **100** may assist the user with completing the process of the function. For example, the function may require the user to operate the vehicle **10** in a particular way or input the values of various function parameters (e.g., to the app). To this end, the app may retrieve the detailed procedure of the function, such as by communication with the one or more servers **140**, and may suggest to the user (e.g., by verbal communication) that the user might receive support to complete the function (e.g., "Would you like assistance with this function?"). After receiving the user's confirmation that assistance is desired, the user may simultaneously instruct the user on the various steps for completing the function while tracking the progress of the process. As needed, the user may converse with the system **100** to ask questions or request additional details, and the system **100** may retrieve function data and any other necessary information from the server(s) **140** with reference to the database(s) **150**. Tracking of the function process may be done automatically by virtue of the connection between the DAT **120** and the vehicle **10**. As conditions remain unmet for completing the function, the app may inform the user of unmet conditions, finally confirming that the function has been done properly. The app may then respond with the result and any further options as described above. In this way, the user may be in communication with the system **100** at all times, to a degree suitable for each particular user, while setting up and performing complex scan tool functions.

[0061] Another extended example showing an iterative application of the disclosed subject matter is shown in the flow chart of FIGS. **8**A and **8**B. The example of FIGS. **8**A and **8**B may represent a specific application of the flow chart of FIG. **7** for a case where the user is to be assisted with driving the vehicle **10** according to specific conditions during performance of a catalyst (CAT) monitoring function. The procedure may begin with the presentation of the drive cycle procedure (e.g., how the vehicle **10** should be driven) to the user as shown. As can be appreciated, it may not be safe to read the procedure while driving. To this end, the system **100** may help read out the procedure step-by-step so that the user can follow along by listening while paying attention to the road. At the same time, the system **100** may monitor completion of each step by checking data received from the vehicle **10** by the DAT **120**, informing the user as needed of met/unmet conditions. As shown in FIGS. **8**A and **8**B, the user may engage in a conversation with the app while the app interfaces with the vehicle **10** via the DAT **120**, utilizing the NLP model **130** and/or server(s) **140** as needed as described above.

[0062] FIG. **9** shows another flow chart describing an example application of the system **100** of FIG. **1** and/or the operational flow of FIG. **2**. In a case where the DAT **120** is always operatively connected to the vehicle **10** (e.g., a dongle that is always plugged in to the OBD port of the vehicle **10**), the system **100** may periodically check the vehicle condition. For example, the app installed on the mobile device **110** may (in accordance with user preferences) autonomously instruct the DAT **120** to regularly perform one or more scan tool functions to check the vehicle condition. The resulting diagnostic data retrieved from the vehicle **10** by the DAT **120** may then be uploaded to the one or more servers **140** for analysis with reference to the one or more diagnostic databases **150**. If this procedure reveals an abnormal diagnostic condition of the vehicle **10**, the app may communicate with the user as described above to inform the user of the abnormal condition and its severity, along with options for further testing (e.g., additional scan tool functions) and/or recommended repairs and parts. In this way, the flow chart of FIG. **9** may represent another way that the flow charts of FIGS. **4**, **5**A/**5**B and **6**A/**6**B may begin, with the system **100** autonomously initiating a dialogue with the user as a problem is discovered, rather than with the user initiating the dialogue as described above. (For example, the flow chart of FIGS. **5**A/**5**B may begin with the system **100** detecting, during a period check, that there is a fault on the engine control module (ECM).) If, on the other hand, the diagnostic data reveals no abnormal condition but only that the vehicle **10** is due (or will soon be due) for regular maintenance, the app may communicate with the user to remind the user and assist the user with scheduling regular maintenance (e.g., which may include directing the user to nearby service centers or presenting contact information of nearby service centers, for example). The system **100** may, for example, read the odometer and remind the user of scheduled maintenance or check the transmission fluid level and inform the user that it is time for a replacement.

[0063] In the example of FIG. **1**, the system **100** is illustrated as including a distinct mobile device **110** and DAT **120**. However, it is contemplated that any or all of the functionality described in relation to the app installed on the mobile device **110** may be combined in a single device that is capable of connecting (by a cable or wirelessly) to the vehicle **10** (e.g., via the OBD port) to perform the scan tool functions described herein, such that the mobile device **110** and DAT **120** may be the same device (e.g., a mobile device **110** with DAT functionality or a DAT **120** with enhanced communication and user interface functionality). Along the same lines, it is contemplated that the NLP model **130** may not necessarily be embodied in a distinct computer and may be resident on the mobile device **110** or DAT **120** in some cases.

[0064] In the above examples, it is described that a pre-trained machine learning model such as a general purpose chatbot or virtual assistant (e.g., OpenAI's ChatGPT) may serve as the NLP model **130**. However, the NLP model **130** need not necessarily be limited in this respect and may, for example, be a scratch-built model that is managed by the same server(s) **140** that have access to the diagnostic database(s) **150**. In such an implementation, the inputs to the model might not be limited to natural language and might include diagnostic data including identifying information of the vehicle **10**, diagnostic trouble codes (DTCs), etc. Such a model may be trained on a combination of natural language with one or more such additional data features that may provide context for the natural language. Thus, when the mobile device **110** uses the model to process the user's communication **12**, instead of providing context in the form of some verbiage (e.g., "in a vehicle") appended to the user's communication **12**, the context may be provided to the model as an additional data feature alongside the user's communication **12**. In a case where a machine learning model is also used downstream to determine which scan tool feature functions are appropriate, it is contemplated that the methodology may be streamlined as desired by using a combined model. For example, the inputs to the model may be the communication **12** plus OBD data-based context features, and the output may be one or more proposed scan tool functions, such that the generation of keywords from the NLP model output may be omitted.

[0065] The present disclosure contemplates an automotive artificial intelligence (AI) assistant system that engages users in an integrated, conversational manner using natural language. It serves as a technical assistant via voice or text for scan tool users, providing professional-level assistance in diagnosing and resolving automotive issues. Implemented across various platforms like the web, smartphones, and scan tools, the AI assistant utilizes vehicle data from a scan tool to offer tailored assistance. The system may suggest failure components or systems, the top three likely fixes, component locations, repair instructions, component inspections, specifics (like wiring color, pin numbers, voltage, etc.), removal and replacement instructions, part number, and/or scheduled maintenance. The automotive AI assistant revolutionizes the user experience by enabling seamless interaction through both voice and text inputs. The dynamic logic is continuously updated based on user feedback, ensuring flexibility and convenience. Distinguishing itself from a simple chatbot, the automotive AI assistant is a mentor and master tech, offering valuable guidance for troubleshooting car problems. By learning from extensive big data, the AI assistant ensures effective assistance in problem-solving for users.

[0066] Users of the disclosed AI assistant can engage in ongoing conversations, leveraging the AI assistant's diagnostic and repair instruction capabilities. The system may employ dynamic logic, continuously updating based on user feedback. As users follow and implement the provided suggestions, the AI assistant may adapt its guidance, refining the diagnostic process. This iterative approach fosters a collaborative environment where users actively participate, completing each suggested step to effectively troubleshoot and resolve automotive issues. The user-centric design ensures a tailored and evolving assistance experience, making the AI assistant an indispensable tool for comprehensive problem-solving in automotive diagnostics.

[0067] FIG. **10** shows an exemplary operational flow of the automotive AI assistant system including the following operations:

Receive Request

[0068] The automotive AI assistant system may be compatible with various platforms such as software, websites, scan tools, pocket smart devices, etc. Users may have the flexibility to submit requests through either voice or text input.

Retrieve Vehicle Data

[0069] Users may be prompted to furnish comprehensive vehicle data, encompassing vehicle identification number (VIN), diagnostic trouble codes (DTC), malfunction indicator lamp (MIL) status, freeze frame, and more. Utilizing the gathered information, the system may then initiate a search and comparison process within the existing data sources to deliver optimal results.

Ask for Symptoms

[0070] Upon acquiring essential information, the system may leverage the data to conduct searches and furnish users with symptoms or indications of vehicle issues as they arise. Subsequently, the application may rely on user confirmation to store and learn the information, thereby enhancing the accuracy of the data.

AI Processing

[0071] Utilizing the supplied information, the application may undergo processing through AI algorithms. It may execute searches across diverse sources, depending on user confirmation. The system may continuously learn and store information, progressively establishing a high-accuracy big data system.

Suggest Assistance

[0072] After collecting sufficient input data and running algorithms, the system may propose a dynamic procedure for user-AI interaction. Referring to FIG. **10**, the AI assistant may utilize inputs like DTC, Live Data, symptoms, and big data to suggest possible failure components/systems (1), providing a list of top verified fixes based on specific big data (2). This quickly informs the user of the faulty area. Additionally, the AI assistant may offer initial inspections for prompt issue resolution (3), especially for intermittent or simple problems like disconnected connectors or broken hoses. The system may provide details on the location of faulty components or systems (4), preparing users for the next steps. After an initial checking, the AI assistant may establish a dynamic logic for a step-by-step repair procedure (5), ensuring components work normally under specific conditions. The procedure also may include the information of harness connector pins, wiring diagrams, or user-guide voltmeter, and others crucial for DIY users. Post-inspection, the AI assistant may suggest the final repair or replacement action, providing OEM part numbers, purchase links, and removal and replacement instructions (6). The system may wait for user confirmation of completed actions, assisting in issue validation (7).

User Confirmation

[0073] The user may confirm whether the AI suggestions are helpful. In the case of unresolved issues, the AI may request additional information for further action.

Completion and Waiting a New Request

[0074] If the problem is resolved, the system may conclude the interaction and await the next user request.

[0075] As illustrated in FIG. **10** and described above, an exemplary method of providing vehicle diagnostics may begin with receiving a request for AI-assisted diagnostics (step **1010**), retrieving diagnostic data obtained from a vehicle (step **1020**), prompting a user to describe one or more symptoms exhibited by the vehicle (step **1030**), and processing the one or more symptoms and the diagnostic data (step **1040**), which may make use of a natural language processing (NLP) model. Exemplary details of these operations are shown in FIGS. **11**A and **11**B. For example, the request contemplated in step **1010** of FIG. **10** may comprise either or both of a manual request (step **1102**) and an auto request (step **1104**) as depicted in FIG. **11**A, which may be received by the scan tool, mobile communication device, server, or other system component that may be responsible for initiating AI processing (step **1106**). A manual request in step **1102** may be, for example, a selection made by a user over a user interface (e.g., via a touchscreen, softkeys, etc.), such as the exemplary "Get Help from Master Tech AI" button, or via natural language processing of the user's spoken or typed command. An auto request in step **1104** may be generated automatically (i.e., without the user making a selection) in response to some event, such as a problem that is detected or predicted based on diagnostic data collected from the vehicle (e.g., by a dongle that remains plugged in to an OBD-II port). In a case where an NLP model is used to process user input (such as the NLP model **130** described in relation to FIG. **1**), keyword(s) may be defined based on the request (step **1108**).

[0076] Contemporaneously with the request (either before, simultaneously with, or after the request), diagnostic data may be retrieved from the vehicle by a scan tool or dongle (step **1110**), such as the DAT **120** shown in FIG. **1**. The diagnostic data may be obtained from the vehicle in response to the request being received, such as in a case where a command is generated for a scan tool or dongle to initiate a scan upon receiving the request. Alternatively, or additionally, the diagnostic data may be obtained from the vehicle at predefined intervals (e.g., by a dongle that is left plugged into the data port). In this case, the request may be generated in response to an evaluation of the diagnostic data as described above. The obtained diagnostic data may be used, together with any information included in the request (e.g., keywords generated using an NLP model) to query one or more servers and/or databases such as the server(s) **140** and diagnostic database(s) **150** described in relation to FIG. **1** (step **1112**). In this way, the contemplated method may search for useful information like feedback, tips, etc., compare a current fix with verified fixes based on feedback history, find issues related to a similar family of vehicles, etc. The search may be specific to the user's vehicle, as may be derived from a VIN included in the diagnostic data, for example. Combining the diagnostic data obtained from the vehicle with vehicle-specific information related to the request and/or the diagnostic data, the operational flow may proceed with analyzing the information using machine learning to create repair instructions and/or other AI assistance for the user (step **1114**).

[0077] In general, a machine learning model may produce the AI assistance based on a variety of inputs. Contemplated machine learning inputs may include, for example, vehicle diagnostic data such as OBD data collected from the vehicle by a scan tool or a dongle (e.g., in communication with a cell phone or other mobile communication device over a short range wireless connection such as a Bluetooth connection), as well as symptomatic data associated with observable characteristics of the vehicle or its operation. Symptomatic data may encompass, for example, the user's reported symptoms, which may be input manually by the user and may include text or speech natural language interpreted by an NLP model such as ChatGPT as described above. Symptomatic data may additionally encompass sensor data of sensors that are arranged to collect symptomatic data such as vibration, temperature, geolocation, image data (e.g., tire tread wear or a scanned VIN), etc., which may be provided by a mobile communication device such as a cell phone (e.g., using an onboard accelerometer, gyroscope, GPS receiver, camera, etc.) and/or by dedicated sensors installed in the vehicle. The machine learning model may evaluate the vehicle diagnostic data in relation to the symptomatic data in order to get a higher resolution fix than would otherwise be possible, such as where there may be multiple possible fixes that are consistent with a DTC but a most likely fix for the combination of the DTC with specific symptomatic data for a particular vehicle. In addition, the machine learning model may determine what additional data may be needed in order to further narrow down the number of possible fixes. In this case, the AI assistance produced by the machine learning model may include guidance to run additional diagnostic tests as described herein.

[0078] As can be appreciated, correctly diagnosing faults in a vehicle's complex systems may often benefit from the ongoing guidance that may be provided by the AI assistance, including guidance based on feedback from the user. For example, the user may initially report a strange sound heard from the vehicle but may not initially recognize that the sound only occurs when the vehicle is turning to the left at 30 miles per hour. Advantageously, the AI processing contemplated by the present disclosure may arrive at the correlation between the reported symptom and the vehicle state much more easily than a human would. For example, the user may simply say, "There's that sound again!" every now and then while driving, and the reported symptom may be ingested by a machine learning model together with the vehicle's live data indicative of the vehicle's state at the moment that the symptom was reported. As a result of having been trained on a large amount of historical data including diagnostic data (DTCs, live data, etc.) as well as symptomatic data (driver-reported data, sensor data, etc.), the machine learning model may immediately recognize a most likely cause of the reported sound that is being heard under the specific vehicle conditions of that particular vehicle (e.g., year, make, model, engine).

[0079] As another example, the machine learning model may recognize that there is a relative lack of reliability in its determination of the cause of some symptom, such as where the same set of input data (e.g., diagnostic data and symptomatic data) that is collected points to two or more likely causes with similar likelihood. In this instance, it may be evaluated that one or more further datapoints would effectively disambiguate between the two or more likely causes by making one far more likely than the other(s). Here, the AI assistance output by the machine learning model may comprise instructions to the user to collect the needed data, for example, by telling the user to operate the vehicle in a particular way or by assisting the user in performing a special function test or otherwise conduct initial testing as described in more detail below. Following the user's additional data collection, the user's feedback of the result may then be added to the machine learning model input in order to get a higher resolution fix.

[0080] As illustrated by these examples, the starting point for the AI assistance may vary depending on the circumstances. In some cases, a driver-reported symptom may initiate the AI assistance, while in other cases the AI assistance may begin in response to some spurious or unexpected signals from sensors in the vehicle or in the driver's mobile device (and/or live data or DTCs collected from an OBD port of the vehicle). In the latter case, the AI processing may begin, unbeknownst to the user, by analyzing any relatedness between such unexpected signals and/or diagnostic data, such as common timing, frequency, signal pattern, etc. to see if they individually or collectively could cause or contribute to the same vehicle condition. Such symptomatic data could come from cellphone sensors, such as an accelerometer (e.g., for vibrations) or an IR sensor connected to the cellphone (e.g., for detection of an abnormal temperature when focused on some potentially malfunctioning component). The AI processing may be performed on the basis of such data, with or without natural language input from a user.

[0081] The contemplated machine learning model(s) may produce output based on a large amount of input data (e.g., sensor data, diagnostic data, etc.), making it possible for the AI processing to recognize or otherwise take into account patterns and correlations that are not discernible to a human. For instance, sensor data from multiple sensors may be automatically combined based on their correlation in timing. As one example, a user may have no intention of taking an infrared (IR) measurement and may be using a cell phone for some other purpose or not using the cell phone at all in the moment the measurement is taken. Data may nevertheless be collected from the IR sensor connected to the cell phone, and the source of the data may simultaneously be determined based on the position and/or orientation of the phone relative to some part of the vehicle (e.g., as may be determined from a gyroscope and/or camera within the phone, possibly making use of an object recognition algorithm). The temperature reading of the vehicle part in question may indicate a part that is hotter or colder than it should be and thus indicative of a malfunction that can be reported to the user. In general, the temperature reading correlated with the determined object or location within the vehicle may with be ingested to the machine learning model and in this way may be automatically correlated with other seemingly unrelated data (such as a driver's spoken inquiry or diagnostic data collected from the OBD port). By taking into account multiple sources of data using machine learning models in this way, the AI processing may provide insights that go beyond those of a human mechanic.

[0082] In some cases, the AI processing may proceed iteratively, with the output of a machine learning model being subsequently used as input to the same or a different machine learning model, which may then produce output having a higher resolution. For example, the driver may say that there is some unusual sound such as a scraping sound. This information may be input to a machine learning model, whose output may indicate some potential causes of the scraping sound. Autonomously, or on demand from the user, the output including the potential causes of the scraping sound may be combined with additional data, such as correlated DTC data and/or live data (e.g., from the same time that the scraping sound was reported). The resulting new output of the machine learning model may provide a higher resolution fix, such as a most likely cause from among the potential causes in light of the additional correlated data. Iterative or recursive exploration of an initial set of potential causes in this way might be requested by the user upon being presented with the potential causes. Alternatively, or additionally, such iterative or recursive exploration might be performed autonomously in response to the set of potential causes being too large (e.g., more than two possible causes) or having a low reliability index (e.g., none of the possible causes is significantly more likely than the others), or based on the availability of correlated data (e.g., the driver has a dongle plugged into the OBD port that was simultaneously collecting data at the time of the request). In an instance where the correlated data is not readily available and requires some further action by the user, the user may be prompted to perform one or more diagnostic tests as described herein in order to narrow down the list of possible causes.

[0083] Advantageously, the machine learning model(s) contemplated herein may be continually trained on the basis of the diagnostic, symptomatic, and/or other data received. For example, when a user implements a repair of the vehicle based on the diagnostic condition determined by the machine learning model, or when the user follows additional diagnostic or repair instructions that are output by the machine learning model, the validity of the determination(s) made by the machine learning model may become known. For example, a user may be prompted to indicate whether the diagnosis was correct, whether the repair was successful, whether the repair instructions were easy to follow, etc. Alternatively, or additionally, the subsequent diagnostic data of the vehicle (e.g., as may be collected from the OBD port) may indicate whether the diagnosis was correct, whether the repair was successful, etc., without any input from the user. By combining feedback of this type from the user and/or from automatically collected diagnostic data with the values of the machine learning model's input and output data that resulted in the diagnosis, repair, or instructions in question, a training data set may be built for further training of the machine learning model. In this way, the machine learning model may improve over time as the determinations it produces are tested.

[0084] Referring back to FIG. **10**, the exemplary method may continue with providing natural language guidance to the user based on a result of the processing (step **1050**). As discussed above, contemplated AI-suggested assistance (which may be the output of a machine learning model as described) may include, for example, possible failure components, likely fixes that have been verified, initial inspection guidance, component location details, step-by-step repair instructions (including wire color, pin number, normal range, etc.), remove and replace instructions, repair validation guidance, etc. Exemplary details of these operations are shown in FIGS. **11**B-**11**D, beginning with the generation of guidelines (step **1116**) that may be the output of the analysis in step **1114**. In general, the guidelines or guidance may comprise one or more step-by-step procedures that may be provided to the user in a conversational natural language format (either as speech or text). The user may or may not be asked to provide feedback or make choices, which may be interpreted by the NLP model based on the user's spoken or typed natural language. The contemplated natural language guidance may include testing instructions for the user to perform a step-by-step procedure to diagnose the vehicle. In the illustrated method, for example, the natural language guidance may begin with initial testing suggested by the AI (step **1118**). Initial testing may comprise relatively quick or easy tests that may help to narrow down the possible issues that are occurring in the vehicle. An example of initial testing is shown in FIG. **11**B, where first the user is asked to check the mass airflow sensor (MAF) connector for damage, then check the engine oil level, etc. The user may report the results at each stage of the initial testing using natural language. Based on the results of initial testing (or in some cases without having conducted initial testing), the natural language guidance may proceed with instructions for most likely fixes suggested by the AI (step **1120** on FIG. **11**C). In contrast to the initial testing, the most likely fixes may represent more involved or difficult procedures. As shown in FIGS. **11**C and **11**D, each most likely fix may be checked one at a time in order to diagnose the vehicle.

[0085] Referring to FIG. **11**C, a procedure for testing a most likely fix may begin with showing safety and preparation requirements (step **1122**), such as a required state of the vehicle, appropriate attire including gloves, etc., after which the user may be given an option to press "start" or otherwise indicate (e.g., using natural language) that he/she is ready to begin receiving instruction. The location of the component or system may then be shown to the user (step **1124**), e.g., on a display screen. The testing procedure may begin with checking whether the component or system in question supports an automated diagnostics function, which may also be referred to as intelligent diagnostics or guided diagnostics, for example, and may enable the component or system to perform a self-test with minimal user input. If such functionality is supported, the procedure may continue with initiating the automated diagnostic routine (step **1128**). Examples of such functionality may be found in the '289 application. In the absence of such automated diagnostics functionality, the procedure may guide the user in testing the most likely fix by showing step-by-step testing instructions and receiving feedback from the user (step **1130**). The testing instructions may include wire colors, pin numbers, harness connector view, expected range values, in order to enable a user with limited knowledge to perform the test correctly. An example is shown in FIG. **11**D, which may begin with instructing the user to first put the engine in an idle state and then use a digital multimeter (DMM) to connect PIN 1 (white) of the MAF sensor with the red lead of the DMM and to connect PIN 2 (blue) of the MAF sensor with the black lead of the DMM, followed by prompting the user to input a measurement value (e.g., a voltage value read by the DMM). The AI may then compare the measurement value with an expected range value (e.g., an expected range of voltages between a particular pin and ground) and suggest the next step of the test based on the result.

[0086] The results of the automated diagnostic routine and/or the manual step-by-step routine performed by the user may then be used to determine whether the component or system in question is good, i.e., functioning correctly (step **1132**). If it is, the system may conclude that the suggested fix was incorrect and that the problem with the vehicle must be due to something else. In this case, the AI may suggest the next most likely fix (step **1134**) and the procedure may repeat, beginning with showing any safety and preparation requirements for the next most likely fix (step **1122**). On the other hand, if the component or system is found to be malfunctioning in step **1132**, the AI suggested assistance of step **1050** (see FIG. **10**) may proceed with providing additional natural language guidance in the form of repair instructions for the user to perform a repair on the vehicle. The repair instructions may include identification information of a replacement part (step **1136**). The user may then perform the repair him/herself or have a service technician perform the repair. It is contemplated that the AI may walk the user through the repair in the same way as the user may be walked through testing the most likely fix, using a natural language dialogue between the user and the system.

[0087] Once it is confirmed that the repair has been performed (step **1138**) (e.g., automatically or through appropriate input by the user), the repair instructions may continue with a verification procedure for confirming that the repair was successful (step **1140**, corresponding to step **1060** in FIG. **10**). For example, the user may be instructed to perform additional tests to verify that the issue has been resolved. If the issue has been successfully resolved (Yes at step **1142**, corresponding to step **1070** in FIG. **10**), the diagnostic test and repair procedure may be completed and the system may await a new request (step **1144**, corresponding to step **1080** in FIG. **10**). If the issue has not been resolved (No at step **1142**), then the procedure may loop back to explore a next most likely fix (step **1134**) as described above. In this case, the system may first gather additional information related to the unsuccessful repair (step **1146**), which may be input by the user, for example.

[0088] The operational flows depicted in FIGS. **10**, **11**, and **11**A-**11**D and, in particular, the AI-assisted communication with the user to diagnose and/or repair a component or system of the vehicle, may be performed by a mobile communication device such as a smart phone or tablet with an installed mobile application, one example of which is the mobile device **110** shown in FIG. **1**. Such a mobile device **110** may retrieve diagnostic data (including a VIN) from a vehicle **10** using a dongle **120** and may communicate with an NLP model **130** and one or more servers **140** connected to a diagnostic database **150** to perform the various AI functionality described above. Alternatively, the role of the mobile device **110** and dongle **120** may be combined in a scan tool. In some instances, the operational flows or portions thereof may be performed by one or more servers **140**, which may communicate directly with an NLP model **130**.

[0089] Owing to the disclosed systems and methods, AI may be used to help technicians by providing a virtual professional mentor, saving them hours of searching the Internet and technical documents to solve problems. The AI may be trained using a database of diagnostic solutions (e.g., built using real vehicle repair reports and documents) and may be auto-updated based on real feedback from a user. The AI may additionally use information retrieved from a DAT such as a scan tool or dongle to assist with a specific user's vehicle, helping to narrow down the problem area and to identify common issues with the particular vehicle in question (which may be known from a VIN included in the diagnostic data obtained from the vehicle). The AI virtual assistant may provide specific information, such as wiring color, signal pin, harness view, and voltage specifications, while avoiding irrelevant information (such as information for a different vehicle). The AI assistant may use vehicle information and retrieved information from the scan tool or dongle (e.g., DTCs and other diagnostic data) to suggest each next step of the diagnostic and/or repair procedure being undertaken. Based on user feedback (natural language text or voice and/or options chosen from a set of AI suggestions), the AI assistant may derive keywords to determine a logical solution for a particular DTC and vehicle. The AI assistant's role may be to analyze the request or feedback from the user, find the keywords that match most with the diagnostic data in a diagnostic database accessed by a server, and find the best solution considering the totality of information. The AI assistant may initiate automated diagnostic routines in addition to utilizing various inputs from the user, for example, by asking the user to perform an active test, a special function test, coding, programming, etc. using a scan tool or mobile app (connected to the vehicle via a dongle) as part of the diagnostic process.

[0090] The functionality described above in relation to the mobile device **110**, DAT **120**, and server(s) **150** shown in FIG. **1**, the disclosed automotive AI assistant system, and the operational flows and flow charts described in relation to FIGS. **2-11**, FIGS. **11**A-**11**D, and throughout the disclosure may be wholly or partly embodied in one or more computers including a processor (e.g. a CPU), a system memory (e.g. RAM), and a hard drive or other secondary storage device. The processor may execute one or more computer programs, which may be tangibly embodied along with an operating system in a computer-readable medium, e.g., the secondary storage device. The operating system and computer programs may be loaded from the secondary storage device into the system memory to be executed by the processor. The computer may further include a network interface for network communication between the computer and external devices (e.g., over the Internet), such as between the mobile device **110** and the DAT **120** or between the mobile device **110** or DAT **120** and the server(s) **140** or third-party computer systems associated with the NLP model **130**. To the extent that any of the described functionality may be performed by the server(s) **140**, the server(s) **140** may comprise multiple physical servers and other computers that communicate with each other to perform the described functionality.

[0091] The above computer programs may comprise program instructions which, when executed by the processor, cause the processor to perform operations in accordance with the various embodiments of the present disclosure. The computer programs may be provided to the secondary storage by or otherwise reside on an external computer-readable medium such as a DVD-ROM, an optical recording medium such as a CD or Blu-ray Disk, a magneto-optic recording medium such as an MO, a semiconductor memory such as an IC card, a tape medium, a mechanically encoded medium such as a punch card, etc. Other examples of computer-readable media that may store programs in relation to the disclosed embodiments include a RAM or hard disk in a server system connected to a communication network such as a dedicated network or the Internet, with the program being provided to the computer via the network. Such program storage media may, in some embodiments, be non-transitory, thus excluding transitory signals per se, such as radio waves or other electromagnetic waves. Examples of program instructions stored on a computer-readable medium may include, in addition to code executable by a processor, state information for execution by programmable circuitry such as a field-programmable gate arrays (FPGA) or programmable logic array (PLA).

[0092] The above description is given by way of example, and not limitation. Given the above disclosure, one skilled in the art could devise variations that are within the scope and spirit of the invention disclosed herein. Further, the various features of the embodiments disclosed herein can be used alone, or in varying combinations with each other and are not intended to be limited to the specific combination described herein. Thus, the scope of the claims is not to be limited by the illustrated embodiments.

What is claimed is:

1. A computer program product comprising one or more non-transitory program storage media on which are stored instructions executable by one or more processors or programmable circuits to perform operations for providing vehicle diagnostics, the operations comprising:

receiving a request for AI-assisted diagnostics;

receiving diagnostic data obtained from a vehicle;

prompting a user to describe one or more symptoms exhibited by the vehicle;

processing the one or more symptoms and the diagnostic data at least in part using a natural language processing (NLP) model; and

providing natural language guidance to the user based on a result of the processing.

2. The computer program product of claim **1**, wherein the diagnostic data is obtained from the vehicle in response to the request being received.

3. The computer program product of claim **2**, wherein the request is derived from speech of the user.

4. The computer program product of claim **2**, wherein the request is derived from text entered by the user.

5. The computer program product of claim **2**, wherein the diagnostic data is obtained autonomously in response to the request being received.

6. The computer program product of claim **1**, wherein the diagnostic data is obtained from the vehicle at predefined intervals, and the request is generated in response to an evaluation of the diagnostic data.

7. The computer program product of claim **6**, wherein the request is generated by a scan tool.

8. The computer program product of claim **1**, wherein the diagnostic data includes identification information of the vehicle.

9. The computer program product of claim **1**, wherein the one or more symptoms are derived from speech of the user.

10. The computer program product of claim **1**, wherein the one or more symptoms are derived from text entered by the user.

11. The computer program product of claim **1**, wherein the natural language guidance includes testing instructions for the user to perform a step-by-step procedure to diagnose the vehicle.

12. The computer program product of claim **11**, wherein the testing instructions include a wire color.

13. The computer program product of claim **11**, wherein the testing instructions include a pin number.

14. The computer program product of claim **11** wherein the testing instructions include an expected range value.

15. The computer program product of claim **11**, wherein the testing instructions include prompting the user to input a measurement value, and the operations further comprise comparing the measurement value to an expected range value.

16. The computer program product of claim **1**, wherein the operations further comprise initiating an automated diagnostic routine based on the result of the processing, wherein the natural language guidance is provided to the user further based on a result of the automated diagnostic routine.

17. The computer program product of claim **1**, wherein the natural language guidance further includes repair instructions for the user to perform a repair on the vehicle.

18. The computer program product of claim **17**, wherein the repair instructions include identification information of a replacement part.

19. The computer program product of claim **17**, wherein the repair instructions include a verification procedure for confirming that the repair was successful.

20. The computer program product of claim **1**, wherein said processing and said providing the natural language guidance proceed autonomously in response to receipt of the one or more symptoms.

21. The computer program product of claim **1**, wherein said processing includes uploading the one or more symptoms and the diagnostic data to one or more servers and receiving the result from the one or more servers.

22. The computer program product of claim **1**, wherein said processing includes inputting the one or more symptoms and the diagnostic data to a machine learning model.

23. The computer program product of claim **22**, wherein said processing includes generating instructions to collect additional data based on an output of the machine learning model.

24. A system comprising:

the computer program product of claim **1**; and

a mobile device, wherein the mobile device includes the one or more processors or programmable circuits.

25. The system of claim **24**, wherein the diagnostic data is obtained from the vehicle by a dongle communicatively coupled with the mobile device.

26. A system comprising:

the computer program product of claim **1**; and

a scan tool, wherein the scan tool includes the one or more processors or programmable circuits and is operable to obtain the diagnostic data from the vehicle.

27. A method of providing vehicle diagnostics, the method comprising:

receiving a request for AI-assisted diagnostics;

receiving diagnostic data obtained from a vehicle;

prompting a user to describe one or more symptoms exhibited by the vehicle;

processing the one or more symptoms and the diagnostic data at least in part using a natural language processing (NLP) model; and

providing natural language guidance to the user based on a result of the processing.

28. A system for providing vehicle diagnostics, the system comprising:

a mobile device operable to receive a request for AI-assisted diagnostics, receive diagnostic data obtained from a vehicle, and prompt a user to describe one or more symptoms exhibited by the vehicle; and

one or more servers in communication with the mobile device, the one or more servers being operable to process the one or more symptoms and the diagnostic data at least in part using a natural language processing (NLP) model;

wherein the mobile device is further operable to provide natural language guidance to the user based on a result of the processing.

29. The system of claim **28**, further comprising a data acquisition and transfer device (DAT) in communication with the mobile device and connected to an OBD port of the vehicle, the DAT being operable to obtain the diagnostic data from the vehicle.

30. A system for providing vehicle diagnostics, the system comprising:

a scan tool operable to receive a request for AI-assisted diagnostics, obtain diagnostic data from a vehicle, and prompt a user to describe one or more symptoms exhibited by the vehicle; and

one or more servers in communication with the scan tool, the one or more servers being operable to process the one or more symptoms and the diagnostic data at least in part using a natural language processing (NLP) model;

wherein the scan tool is further operable to provide natural language guidance to the user based on a result of the processing.

31. A computer program product comprising one or more non-transitory program storage media on which are stored instructions executable by one or more processors or programmable circuits to perform operations for providing vehicle diagnostics, the operations comprising:

receiving diagnostic data obtained from a vehicle;

receiving symptomatic data obtained from one or more sensors within the vehicle;

inputting the diagnostic data and the symptomatic data to a machine learning model; and

providing natural language guidance to the user based on an output of the machine learning model.

32. The computer program product of claim **31**, wherein the operations further comprise building a training data set based on the diagnostic data, the symptomatic data, and the output of the machine learning model.

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